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Instituto de Ciências Matemáticas e de Computação

Asymptotics of Partition Functions of Coulomb Gases

João Victor Alcântara Pimenta

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João Victor Alcântara Pimenta

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Concentration Area: Mathematics

Advisor: Prof. Dr. Guilherme Lima Ferreira Silva

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"En remontant chez moi pour y passer la soirée à travailler de mon mieux,
je me disais que le monde n'est pas construit pour l'équilibre.
Le monde est désordre.
L'équilibre n'est pas la règle, c'est l'exception.
Et je faisais le serment de travailler pour l'ordre et l'équilibre.
Que de serments! Tu vas sourire."
G.Duhamel, Maitres, 1937

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For all of those who were before and are now:
Your courage, patience, daring, and caring has brought me here.
To hoping that I become deserving of such act, for I feel loved and dear; of immense luck.
For all of those who will be after:
To promise that I can for you as it was done for me.

"João, é bom ver que você aprendeu e fez tantas coisas bonitas!
Continue assim, muito feliz! Brinque e aprenda muito."
Te amo muito, Papai.

RESUMO

JOÃO V. A. PIMENTA. **Assintóticas de Funções Partição em Gases de Coulomb**. 2026. 65 p. Dissertação (Mestrado em Ciências – Matemática) – Instituto de Ciências Matemáticas e de Computação, Universidade de São Paulo, São Carlos – SP, 2026.

As matrizes aleatórias podem ser vistas de duas formas: como uma função mensurável de um espaço de probabilidade num conjunto de matrizes ou, mais concretamente, como uma matriz cujas entradas são variáveis aleatórias com alguma distribuição conjunta. Um conjunto de tais matrizes com uma dada distribuição arbitrária é designado por ensemble. Para ensembles de interesse, pode-se estudar estatísticas como autovetores e distribuições de autovalores. As simetrias do conjunto escolhido irão refletir sobre quais modelos tais estatísticas podem representar. Por exemplo, as distâncias dos valores característicos de uma matriz simétrica com coeficientes aleatórios modelam as distâncias entre níveis sucessivos de energias de núcleos atômicos pesados. Mais interessante ainda, existe uma certa universalidade nas estatísticas encontradas entre classes de ensembles, sugerindo a relevância da teoria no estudo da universalidade nos fenómenos modelados em geral. Particularmente importante é o estudo de tais matrizes quando estas crescem infinitamente, considerando assintoticamente o seu comportamento estatístico. O nosso foco é uma técnica para derivar assintóticas para a chamada função partição de alguns ensembles interessantes de matrizes aleatórias.

Palavras-chave: Teoria de Matrizes Aleatórias, Ensembles de Matrizes Planares, Assintótica.

ABSTRACT

JOÃO V. A. PIMENTA. **Asymptotics of Partition Functions of Coulomb Gases** . 2026. 65 p. Dissertação (Mestrado em Ciências – Matemática) – Instituto de Ciências Matemáticas e de Computação, Universidade de São Paulo, São Carlos – SP, 2026.

Random Matrices can be though of in two ways: As a measurable function from a probability space into a set of matrices or, more concretely, as a matrix whose entries are random variables with some joint distribution. A set of such matrices with a given arbitrary distribution is called an ensemble. One can, for ensembles of interest, study statistic such as the eigenvectors and eigenvalue distributions. The symmetries of chosen set will reflect on what models such statistics can model. For example, distances of the characteristic values of a symmetric matrix with random coefficients model distances between successive levels of energies of heavy nuclei. More interestingly, there is some universality in the statistics found among classes of ensembles, suggesting the relevance of the theory in studying the universality in general modeled phenomena. Particularly important is the study of such matrices when they grow infinitely large, taking asymptotically its statistical behavior. We focus on a technique for deriving asymptotics for the so-called partition function of some interesting ensembles of random matrices.

Keywords: Random Matrix Theory, Planar Matrix Ensembles, Asymptotics.

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PLANAR ENSEMBLES

We are going to discuss in this chapter the ideas developed in the paper Byun, Kang and Seo (2023), where asymptotics were develop for a large class of random matrix ensembles using relatively straightforward asymptotic techniques taking advantage of the orthogonal polynomials present in these models. In the end, we hope to clarify the origin for the novel limit-behavior expression for the *Partition Function* $\mathcal{Z}_N^{\beta=2}$. Namely, for the expression

$$\ln(\mathcal{Z}_{N,\beta=2}^{Q}) = -I_Q N^2 + \frac{1}{2} N \ln N + E_Q N + G \ln N + F^Q + o(1). \tag{6.1}$$

Partition functions perform the role of the normalization for the probability measure relative to the Coulomb gas being worked on - a system of interacting particles confined by some potential. In this instance, we deal with complex plane gases, parameterized by the variable $\beta = 2$. This relates in RMT, to the famous *random normal matrix* and *planar symplectic ensembles*, depending on the symmetry imposed. If one takes, for example, the quadratic potential $Q(x) = x^2/2$ keeping the $2-\beta$ regime, one gets the *complex* and *planar* Ginibre ensembles, falling back into the classical models. For the results, another restriction is made, the confining potentials must be radially symmetric. For a large class of potentials Q, the first term I_Q was known for a long time, being given by the energy of the associated equilibrium measure. The coefficient $1/2$ of the order $N \ln N$, as well as the entropy term E_Q, were known from recent work of Leblè and Serfaty (2017), also for rather large classes of Q. The remaining terms in this expansion are new. The term G appears to depend only on whether the limiting spectrum is an annulus or a disk which, surprisingly, seems to not have been observed in the previous literature.

Although some recent work such as in Hedenmalm and Wennman (2020) may indicate the direction to expand some of these results to larger classes of potentials, this is, so far, still a necessary condition for the techniques presented henceforth. Once the model is established, the techniques are involved, but should be reasonable: We express the partition function in terms of a summation of norming constants, for which we have a integral representation. Then, the integral can be estimated using the Laplace technique - separating the integral in a central main-contributing region and a bounded error-contributing complement region. Finally, the limit on the summation can be estimated with quadrature techniques, namely the Euler-Maclaurin formula, giving us the final results after computations.

6.1 Working Model

In this section we properly define the model and specifics of the working model to which we wish to get the asymptotics from. We have discussed in **??** how eigenvalue statistics can be interpret as some appropriate Coulomb gas. On the complex plane one could have for example the measure

$$\mathrm{dp}_N^{(\beta)}(z_1,\cdots,z_N) := \frac{1}{\mathcal{Z}_N^{(\beta)}} \prod_{j>k=1}^{N} |z_j - z_k|^\beta \prod_{j=1}^{N} \mathrm{e}^{-\frac{\beta N}{2}Q(z_j)}\, \mathrm{d}A(z_j) \tag{6.2}$$

where $\mathrm{d}A(z) := \mathrm{d}^2 z/\pi$ is the area measure. This measure would imply that the normalization constant - or in our context, the *Partition Function*, is given by

$$\mathcal{Z}_N^{(\beta)} := \int_{\mathbb{C}^N} \prod_{j>k=1}^{N} |z_j - z_k|^\beta \prod_{j=1}^{N} \mathrm{e}^{-\frac{\beta N}{2}Q(z_j)}\, \mathrm{d}A(z_j). \tag{6.3}$$

Definition 1. *Let $\mathrm{p}_N^{(\beta)}$ be as in Equation (6.2). If $\beta = 2$, $\mathrm{p}_N^{(2)}$ is the measure for the ensemble we call **random normal matrix**. If, for example, one sets $Q(z) = |z|^2$ this would be called **complex Ginibre ensemble**.*

We could also define the measure for the configurational canonical Coulomb gas ensemble in the upper-half plane - which has an additional complex conjugation symmetry and is governed by the law

$$\mathrm{d}\tilde{\mathrm{p}}_N^{(\beta)}(z_1,\cdots,z_N) := \frac{1}{\mathcal{Z}_N^{(\tilde{b}eta)}} \prod_{j>k=1}^{N} |z_j - z_k|^\beta |z_j - \bar{z}_k|^\beta \prod_{j=1}^{N} |z_j - \bar{z}_j|^\beta \mathrm{e}^{-\frac{\beta N}{2}Q(z_j)}\, \mathrm{d}A(z_j) \tag{6.4}$$

and would imply the normalization constant

$$\mathcal{Z}_N^{(\tilde{b}eta)} := \int_{\mathbb{C}^N} \prod_{j>k=1}^{N} |z_j - z_k|^\beta |z_j - \bar{z}_k|^\beta \prod_{j=1}^{N} |z_j - \bar{z}_j|^\beta \mathrm{e}^{-\frac{\beta N}{2}Q(z_j)}\, \mathrm{d}A(z_j). \tag{6.5}$$

Definition 2. *Let $\mathrm{p}_N^{(\beta)}$ be as in Equation (6.4). If $\beta = 2$, $\mathrm{p}_N^{(2)}$ is the measure for the ensemble we call **planar symplectic ensemble**. If, for example, one sets $Q(z) = |z|^2$ this would be called **symplectic Ginibre ensemble**.*

Virtually all results we'll state for the random normal matrix ensemble are valid in the context of the planar symplectic ensemble with only a few adaptions. However, in the name of history telling and simplicity we will omit this measure: Adding this case to our chapter would cloud the point. For now on, we will only consider $\beta = 2$, this has some algebraic advantages that become clear by the Determinantal Point Processes structure (see Chapter 3 for details) - because of this, we omit the β in the variables ($\mathcal{Z}_N^{(\beta)} := \mathcal{Z}_N$).

For the models of random matrices we are interested in, we must consider $\{P_n(z) = \sum_{j=0}^{n} a_j z^j\}_n^{(w)}$ a family of orthogonal polynomials in respect to w where P_m has order m. We're dealing with weight functions of the type

$$w(z) = \mathrm{e}^{-NQ(z)}, \tag{6.6}$$

where $Q(z)$ is some complex potential that satisfies:

1. We have
$$Q(z) = q(|z|) \tag{6.7}$$
for some $q : \mathbb{R} \mapsto \mathbb{R}$, i.e., Q is a radially symmetric potential;

2. The potential Q grows sufficiently fast, that is,
$$\liminf_{|z|\to\infty} \frac{Q(z)}{2\ln(|z|)} > 1; \tag{6.8}$$

3. Furthermore, assume Q to be smooth, subharmonic in $\mathbb{C}$, and strictly subharmonic in a neighborhood of the droplet.

This demands are essential for what follow and not at all counter-intuitive. to require the radial symmetric is to say that the integral in the plane can be somehow exchanged by a line integral as all directions would be homogeneous. The growth requirement comes from the need for a confinement of the particles in the system. Remember that the interaction energy has logarithmic behavior. Smoothness makes sure we can expand the potential in a series without much worries about the continuity and existence of derivatives. The conditions on subharmonicity may not be as clear but are requirements that can be though of as convexity. This will ensure uniqueness of the minima for the function and the connectedness of the support of the equilibrium measure. There is still a question of what are exactly the orthogonal polynomials for such weight function.

Lemma 1. *The set $\{z^j\}_j^{(w)}$ is the proper orthogonal family in relation to the weight function $w(z) = \mathrm{e}^{-NQ(z)}$ given.*

Proof. We know by the assumption of orthogonality that

$$\langle P_n, P_m \rangle_w = \begin{cases} 0 & \text{if } m = n, \\ 1 & \text{if } m \neq n. \end{cases} \tag{6.9}$$

Let us take some arbitrary base of the polynomial space $\{z^j\}_j^{(w)}$. We can apply the Gram Schmidt scheme to get the proper orthogonality in respect to the weight w. If $P_n = x^n$, then the orthogonal element of the basis $\tilde{P}_n$ should be

$$\tilde{P}_n = P_n - \sum_{j=0}^{n-1} \frac{\langle P_n, P_j \rangle_w}{\langle P_j, P_j, \rangle_w} P_j, \tag{6.10}$$

but notice that

$$\begin{aligned}
\langle P_n, P_j \rangle_w &= \int_{\mathbb{C}} z^{n-j} \, \mathrm{e}^{-NQ(z)} \, \mathrm{d}A(z), & (6.11)\\
&= \int_0^\infty \int_0^{2\pi} r^{n-j} \, \mathrm{e}^{\mathrm{i}\theta(n-j)} \, \mathrm{e}^{-Nq(r)} \, r \mathrm{d}r \mathrm{d}\theta, & (6.12)\\
&= \int_0^{2\pi} \mathrm{e}^{\mathrm{i}\theta(n-j)} \, \mathrm{d}\theta \int_0^\infty r^{n-j+1} \, \mathrm{e}^{-Nq(r)} \, \mathrm{d}r, & (6.13)\\
&= 0, \quad \text{since } n \neq j. & (6.14)
\end{aligned}$$

Therefore, we have that $\{z^j\}_j^{(w)}$ is the proper orthogonal family. More than that, if we impose the normality,

$$\int_{\mathbb{C}} a_n^2 z^{n-n} \, \mathrm{e}^{-NQ(z)} \, \mathrm{d}A(z) = \int_{\mathbb{C}} a_n^2 \, \mathrm{e}^{-NQ(z)} \, \mathrm{d}A(z) = 1 \tag{6.15}$$

$$\implies a_n = \sqrt{\frac{1}{\int_{\mathbb{C}} w(z) \mathrm{d}A(z)}}. \tag{6.16}$$

□

6.2 Asymptotics on Droplets

We will use some radially symmetric potential $Q(z) : \mathbb{C} \to \mathbb{R}$ which satisfies some nice growth and smoothness conditions, as specified in section 6.1. Because of the radial symmetry, one can take some function $q : \mathbb{R} \to \mathbb{R}$ such that $q(|z|) = Q(z)$; If we can say that the Q is subharmonic on the plane, i.e., $\Delta Q(z) > 0$ for all $z \in \mathbb{C}$, then

$$\Delta Q(z) \geq 0 = \frac{1}{r}(rq'(r))' \geq 0 \implies rq'(r) \text{ is increasing } \forall r \in \mathbb{R}. \tag{6.17}$$

and strictly increasing in the droplet by the same reason. We begin by quickly considering the auxiliary result.

Lemma 2. *When dealing with a p.d.f. of invariant measure, one can write*

$$\mathcal{Z}_n = n! \prod_{k=0}^{n-1} r_k, \tag{6.18}$$

where r_k is the norming constant as in $\langle q_j^{(2)}, q_k^{(2)} \rangle^{(2)} = r_j^{(2)} \chi_{j=k}$.

Proof. We start stating the following equality which follows from **??** that

$$\det\left[K_n^{(2)}(\lambda_i,\lambda_j)\right]_{i,j=1}^n = p^{(w)}(\lambda_1,\lambda_2,\dots,\lambda_n;\beta), \tag{6.19}$$

$$\prod_{k=1}^n r_k^{-1}\det\left[\sqrt{w(x)w(y)}\sum_{k=0}^{N-1} q_k^{(2)}(x)q_k^{(2)}(y)\right]_{i,j=1}^n = \prod_{1<i\leq j<n}(\lambda_j-\lambda_i)^2\prod_{k=0}^n w(\lambda_k) \tag{6.20}$$

therefore

$$n!\det\left[\tilde{K}_n^{(2)}(\lambda_i,\lambda_j)\right]_{i,j=1}^n = n!\prod_{k=1}^n r_k\prod_{1<i\leq j<n}(\lambda_j-\lambda_i)^2\prod_{k=0}^n w(\lambda_k) \tag{6.21}$$

where

$$\mathcal{Z}_{n,2} = n!\prod_{k=1}^n r_k \tag{6.22}$$

□

Thus, one can write the free energy as

$$\ln\mathcal{Z}_N = \ln N! + \sum_{j=0}^{N-1}\ln h_j, \quad \text{where} \qquad h_k := \int_{\mathbb{C}}|z|^{2j}\,\mathrm{e}^{-NQ(z)}\,\mathrm{d}A(z) \tag{6.23}$$

are the norming constant for the monomial family of orthogonal polynomials in respect to w as in Equation 6.6. One could also write, for the planar symplectic ensemble

$$\ln\tilde{\mathcal{Z}}_N = \ln N! + \sum_{j=0}^{N-1}\ln 2\tilde{h}_{2j+1}, \quad \text{where} \qquad \tilde{h}_k := \int_{\mathbb{C}}|z|^{2j}\,\mathrm{e}^{-2NQ(z)}\,\mathrm{d}A(z). \tag{6.24}$$

The expressions derived for the normal ensemble would be easily adapted for this model and it is done in Byun, Kang and Seo (2023), however we will not follow that computation in name of clarity and storytelling.

In order to derive the asymptotic formulas for the free energy we need to derive the asymptotics for the orthogonal polynomials, specifically for the integral of its norming constants expressed in Equation 6.23. The first step in the asymptotics is then to prepare the integral for the Laplace method. Our goal is to identify a region that gives the main contribution, in order, for value of the integral - and consequently to identify a error-contributing region of the integration domain. Let's reformulate our expression to identify the usual integral form for the Laplace method, such as in Chapter 4. We know $Q(z) = q(|z|)$ a radially symmetric potential such that, defining $z := r\,\mathrm{e}^{\mathrm{i}\theta}$, that is, $|z| = r$, we would have

$$h_j = \int_\Theta\int_{\mathbb{R}_+} r^{2j}\,\mathrm{e}^{-Nq(r)}\,\frac{r}{\pi}\mathrm{d}r\mathrm{d}\theta \tag{6.25}$$

$$= \frac{2\pi}{\pi}\int_{\mathbb{R}_+} r^{2j+1}\,\mathrm{e}^{-Nq(r)}\,\mathrm{d}r \tag{6.26}$$

$$= \int_{\mathbb{R}_+} 2r\,\mathrm{e}^{-N\left(q(r)-\frac{2j}{N}\ln(r)\right)}\,\mathrm{d}r. \tag{6.27}$$

We first redefine our potential such that the integrand has a expression better suited to apply the Laplace method. Using

$$V_{\tau(j)} = q(r) - 2\tau(j)\ln(r), \tag{6.28}$$

where $\tau(j) := j/N \in [0,1)$, then one can rewrite the integral as

$$h_j = \int_{\mathbb{R}_+} 2r\, \mathrm{e}^{-NV_{\tau(j)}(r)}\, \mathrm{d}r. \tag{6.29}$$

Now, we have set the stage for the Laplace technique. Note that now τ will perform the function of the previous index j and relates to which coefficient we're calculating. Taking a look at the first derivative of the newly defined potential we find that the critical point r_c must satisfy

$$V'_\tau(r) = q'(r) - \frac{2\tau}{r} \quad \implies \quad r_c q'(r_c) = 2\tau. \tag{6.30}$$

This is: We can find a critical point of the potential r_c, parameterized by the degree of the polynomial we are calculating, or more concretely, by τ. Of course, because we know the range of τ, it is worth defining the limital points

$$\begin{cases} r_0, & \text{the largest solution of } r_0 q'(r_0) = 0 \\ r_1, & \text{the smallest solution of } r_1 q'(r_1) = 2 \end{cases} \tag{6.31}$$

Our main goal here is to evaluate the integral using the surroundings of the critical point. If this point is unique, then we need no more than to calculate the integral in a single small interval around $r_c(\tau)$. This is, of course, true: We have mentioned that the properties on the potential directly implies that $rq'(r)$ is a strictly increasing function on $[r_0, r_1]$, then the solution must be unique. Now, as outside the droplet the growth is not strict, the definitions of r_0 ad r_1 had to be as mentioned before: Respectively the largest and smallest solution to the proper instance of $rq'(r) = 2\tau$. An important distinction must be made between the cases when $r_0 = 0$ and $r_0 > 0$, corresponding to subsection 6.2.2 and subsection 6.2.1, respectively. We begin noticing that $r_\tau | r_\tau q'(r_\tau) = 2\tau$ is an increasing function on τ and that $r_0 \le r_\tau \le r_1$ $\forall \tau \in [0,2]$. As we'll be performing the Laplace technique, Gaussian integrals will be a big part of the result ,more details on why can be found in Chapter 4. As such, we state the following proposition.

Proposition 1. *The Gaussian integrals below can be evaluated as follows.*

$$\int_{\mathbb{R}} \mathrm{e}^{-2ar^2}\, \mathrm{d}r = \sqrt{\frac{\pi}{2}}\frac{1}{a^{1/2}} \quad \implies \quad \int_{\mathbb{R}} \mathrm{e}^{-2\left(\frac{V^{(2)}_{\tau(j)}(r_{\tau(j)})}{4}\right)y^2} = \sqrt{\frac{\pi}{2}}\left(\frac{4}{V^{(2)}_{\tau(j)}(r_{\tau(j)})}\right)^{\frac{1}{2}}, \tag{6.32}$$

$$\int_{\mathbb{R}} \mathrm{e}^{-2ar^2} r^4 \mathrm{d}r = \sqrt{\frac{\pi}{2}}\frac{3}{16}\frac{1}{a^{5/2}} \quad \implies \quad \int_{\mathbb{R}} \mathrm{e}^{-2\left(\frac{V^{(2)}_{\tau(j)}(r_{\tau(j)})}{4}\right)y^2} y^4 = \sqrt{\frac{\pi}{2}}\frac{3}{16}\left(\frac{4}{V^{(2)}_{\tau(j)}(r_{\tau(j)})}\right)^{\frac{5}{2}}, \tag{6.33}$$

$$\int_{\mathbb{R}} \mathrm{e}^{-2ar^2} r^6 \mathrm{d}r = \sqrt{\frac{\pi}{2}}\frac{15}{64}\frac{1}{a^{7/2}} \implies \int_{\mathbb{R}} \mathrm{e}^{-2\left(\frac{V^{(2)}_{\tau(j)}(r_{\tau(j)})}{4}\right)y^2} y^6 = \sqrt{\frac{\pi}{2}}\frac{15}{64}\left(\frac{4}{V^{(2)}_{\tau(j)}(r_{\tau(j)})}\right)^{\frac{7}{2}}. \tag{6.34}$$

6.2.1 Asymptotics - Annulus

We start by considering the annulus regime, with $r_0 > 0$. We know the measure in Equation 6.2 to be the law for the distribution of the particles in this problem. When we say that $r_0 > 0$, we are saying that the region of space $[r_0, r_1]$ where this measure is non-zero for $N \to \infty$ has the annulus shape. If we think of the equivalent Coulomb gas, this means the particles will concentrate in this shape when we perform an asymptotic on N. This is the support for the limit measure.

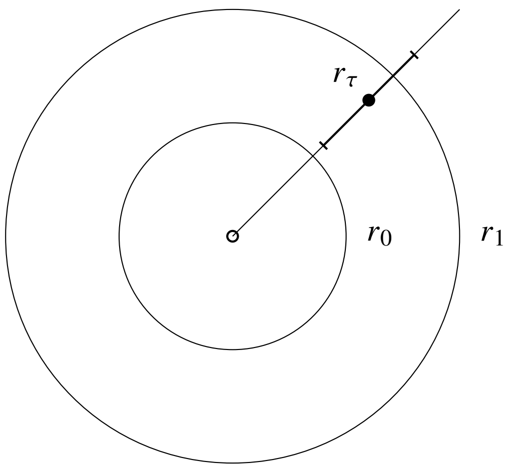

Figure 1 – The figure depicts an annulus referring to the equilibrium measure when $r_0 > 0$. As the complex plane integral is reduced to a line integral, we draw the domain change and the point of interest r_τ where V has a critical point. Around this point we depict the region around the minima, which in the method that should account for the main order of the integral we solve for.

Theorem 1. *For the annulus case, where $r_0 > 0$, we should have the following asymptotic for the free energy asymptotically on N given by*

$$\ln \mathcal{Z}_N = -N^2 I_Q[\mu_Q] + \frac{N}{2}\left(\ln(8\pi) - E_Q[\mu_Q] - 2\right) + \frac{N}{2}\ln(N) + \frac{\ln(2\pi)}{2} + \frac{1}{2}\ln(N) + F_V[A_{[r_0,r_1]}] + O(N^{-1}) \tag{6.35}$$

where we set

$$F_V[A_{[r_0,r_1]}] := -\frac{2}{3}\ln\left(\frac{r_1}{r_0}\right) + \frac{5}{32}\ln\left(\frac{V_1^{(2)}(r_1)}{V_0^{(2)}(r_0)}\right) - \frac{1}{32}\left(\frac{r_1 V_1^{(3)}(r_1)}{V_1^{(2)}(r_1)} - \frac{r_0 V_0^{(3)}(r_0)}{V_0^{(2)}(r_0)}\right) - \frac{1}{12}\int_{r_0}^{r_1} \frac{V^{(3)}(r)^2}{V^{(2)}(r)^2} r \mathrm{d}r, \tag{6.36}$$

$$I_Q[\mu_Q] := \left[\frac{1}{4}\int_{r_0}^{r_1} q(s)sV^{(2)}(s)\mathrm{d}s - \ln(r_1) + \frac{1}{2}q(r_1)\right], \tag{6.37}$$

and

$$E_Q[\mu_Q] := \frac{1}{2}\int_{r_0}^{r_1} \ln(V^{(2)}(s))(sq'(s))'\mathrm{d}s. \tag{6.38}$$

We start by studying the asymptotic for the norming constants h_j. Then, we try to derive the formulas for the summation on its logarithm $\ln(h_j)$. These results should be enough to prove the previous theorem. The following subsections are referring to these two results.

6.2.1.1 Asymptotic for the norming constant

Theorem 2. *We have the following asymptotic on N for h_j, with $0 \le j \le N-1$,*

$$h_j = \frac{\mathrm{e}^{-NV_{\tau(j)}(r_{\tau(j)})}}{\sqrt{N}}\left(\frac{8\pi r_{\tau(j)}^2}{V_{\tau(j)}^{(2)}(r_{\tau(j)})}\right)^{\frac{1}{2}}\left[1 + \frac{1}{N}\mathfrak{B}(r_{\tau(j)}) + O(N^{-2})\right]. \tag{6.39}$$

where we have set

$$\mathfrak{B}(r_{\tau(j)}) := -\frac{V_{\tau(j)}^{(4)}(r_{\tau(j)})}{V_{\tau(j)}^{(2)}(r_{\tau(j)})^2}\frac{1}{8} - \frac{V_{\tau(j)}^{(3)}(r_{\tau(j)})}{V_{\tau(j)}^{(2)}(r_{\tau(j)})^2}\frac{1}{2r_{\tau(j)}} - \frac{V_{\tau(j)}^{(3)}(r_{\tau(j)})^2}{V_{\tau(j)}^{(2)}(r_{\tau(j)})^3}\frac{5}{24}, \tag{6.40}$$

The proof for such theorem passes trough Lemmas 3 and 4. We begin as usual when applying asymptotic techniques in integrals; separating a central part where the integral value is concentrated and another part where we can control the order of contribution. This is possible because the integrand is proportional to some exponential, with negative exponent, with a unique critical point, around which, the value of the expression concentrates, as discussed in Chapter 4. Define $\delta_N = \ln(N)/\sqrt{N}$ - this will be our the cut-size for the interest region and will be justified during the calculations - we have

$$h_j = \int_{|r-r_j|>\delta_N} 2r\,\mathrm{e}^{-NV_{\tau(j)}(r)}\,\mathrm{d}r + \int_{|r-r_j|<\delta_N} 2r\,\mathrm{e}^{-NV_{\tau(j)}(r)}\,\mathrm{d}r = (1) + (2). \tag{6.41}$$

As mentioned, the first integral must be somehow controlled and the second estimated as the main value for the result. We begin by showing that the first term has an error contribution.

Lemma 3. *If we choose M to be such that*

$$\int_{|r-r_j|>\delta_N} \left(\mathrm{e}^{-q(r)}r^2\right)^M r\mathrm{d}r < \infty, \tag{6.42}$$

then, with (1) as in Equation 6.41 and $0 \le j \le N-1$, asymptotically on N,

$$(1) := \int_{|r-r_j|>\delta_N} 2r\,\mathrm{e}^{-NV_{\tau(j)}(r)}\,\mathrm{d}r = \mathrm{e}^{-NV_{\tau(j)}(r_{\tau(j)})}\,\epsilon_N \tag{6.43}$$

where $\epsilon_N = O(\mathrm{e}^{-c(\ln(N))^2})$.

Proof. Let's first show that we can control the first part of the integral, for that, rewrite

$$(1) = \mathrm{e}^{-NV_{\tau(j)}(r_{\tau(j)})} \int_{|r-r_j|>\delta_N} 2r\,\mathrm{e}^{-N\left(V_{\tau(j)}(r)-V_{\tau(j)}(r_{\tau(j)})\right)}\,\mathrm{d}r, \tag{6.44}$$

$$= \mathrm{e}^{-NV_{\tau(j)}(r_{\tau(j)})} \int_{|r-r_j|>\delta_N} 2r\,\mathrm{e}^{-N\left(V_{\tau(j)}(r)-V_{\tau(j)}(r_{\tau(j)})\right)}\,\mathrm{e}^{-(M-M)\left(V_{\tau(j)}(r)-V_{\tau(j)}(r_{\tau(j)})\right)}\,\mathrm{d}r, \tag{6.45}$$

$$= \mathrm{e}^{-NV_{\tau(j)}(r_{\tau(j)})} \int_{|r-r_j|>\delta_N} 2r\,\mathrm{e}^{-M\left(V_{\tau(j)}(r)-V_{\tau(j)}(r_{\tau(j)})\right)}\,\mathrm{e}^{-(N-M)\left(V_{\tau(j)}(r)-V_{\tau(j)}(r_{\tau(j)})\right)}\,\mathrm{d}r. \tag{6.46}$$

Now, we need some way to estimate the difference $(V_{\tau(j)}(r) - V_{\tau(j)}(r_{\tau(j)})$. If we are able to say that this difference is at least $O(N^{-1})$, we could control the growth of the integral choosing M to be large enough such that

$$\int_{|r-r_j|>\delta_N} \left(\mathrm{e}^{-q(r)}\, r^{2\tau(j)}\right)^M r\mathrm{d}r \tag{6.47}$$

converges. Using the definition of the function V and the mean value theorem, we start by evaluating the difference $\Delta(j) = V_{\tau(j+1)}(r_{\tau(j+1)}) - V_{\tau(j)}(r_{\tau(j)})$ in the following manner

$$\Delta(j) = q(r_{\tau(j+1)}) - q(r_{\tau(j)}) - 2\left(\tau(j+1)\ln(r_{\tau(j+1)}) - \tau(j)\ln(r_{\tau(j)})\right), \tag{6.48}$$

$$= q'(c_1)(r_{\tau(j+1)} - r_{\tau(j)}) - \frac{2}{N}\left(\ln(r_{\tau(j+1)}) + j\ln\left(\frac{r_{\tau(j+1)}}{r_{\tau(j)}}\right)\right), \tag{6.49}$$

$$= q'(c_1)r'(c_2)(\tau(j+1) - \tau(j)) - \frac{2}{N}\left(\ln(r_{\tau(j+1)}) + jO(N^{-1})\right), \tag{6.50}$$

$$= q'(c_1)r'(c_2)O(N^{-1}) - O(N^{-1}), \tag{6.51}$$

$$= O(N^{-1}). \tag{6.52}$$

The last equality holds as both functions are continuous and c_1, c_2 are values that are limited; in a compact disk on the plane. Then, we can add zero and use the estimation

$$V_{\tau(j+1)}(r_{\tau(j+1)}) - V_{\tau(j+1)}(r) + V_{\tau(j+1)}(r) - V_{\tau(j)}(r_{\tau(j)}) = O(N^{-1}) \tag{6.53}$$

to derive the result we are after. It is enough to estimate $V_{\tau(j+1)}(r) - V_{\tau(j)}(r_{\tau(j)})$. For that, because $rq'(r)$ is increasing, for all $r > r_{\tau(j)} + \delta_N$ we can say that

$$V_{\tau(j+1)}(r) \geq V_{\tau(j+1)}(r_{\tau(j)} + \delta_N), \tag{6.54}$$

$$= V_{\tau(j+1)}(r_{\tau(j+1)}) + \frac{1}{2}V''_{\tau(j+1)}(r_{\tau(j+1)})(r_{\tau(j)} + \delta_N - r_{\tau(j+1)})^2 + O(\delta_N^3), \tag{6.55}$$

and therefore can write

$$V_{\tau(j+1)}(r) - V_{\tau(j)}(r_{\tau(j)}) = \frac{1}{2}V''_{\tau(j+1)}(r_{\tau(j+1)})(r_{\tau(j)} + \delta_N - r_{\tau(j+1)})^2 + O(\delta_N^3), \tag{6.56}$$

$$= \delta^2 c + O(N^{-1}) + O(\delta_N^3), \tag{6.57}$$

$$\geq \delta^2 c. \tag{6.58}$$

Finally, using this one can say

$$V_{\tau(j+1)}(r) - V_{\tau(j+1)}(r_{\tau(j+1)}) \geq c\delta_N^2. \tag{6.59}$$

Getting this inequality back to the integral expression end up with

$$(1) = \mathrm{e}^{-NV_{\tau(j)}(r_{\tau(j)})} \int_{|r-r_j|>\delta_N} 2r\,\mathrm{e}^{-M\left(V_{\tau(j)}(r)-V_{\tau(j)}(r_{\tau(j)})\right)}\,\mathrm{e}^{-(N-M)\left(V_{\tau(j)}(r)-V_{\tau(j)}(r_{\tau(j)})\right)}\,\mathrm{d}r, \tag{6.60}$$

$$\leq \mathrm{e}^{-NV_{\tau(j)}(r_{\tau(j)})}\,\mathrm{e}^{-c\delta_N^2(N-M)} \int_{|r-r_j|>\delta_N} 2r\,\mathrm{e}^{-M\left(V_{\tau(j)}(r)-V_{\tau(j)}(r_{\tau(j)})\right)}\,\mathrm{d}r, \tag{6.61}$$

where you can check that

$$\int_{|r-r_j|>\delta_N} 2r\,\mathrm{e}^{-M\left(V_{\tau(j)}(r)-V_{\tau(j)}(r_{\tau(j)})\right)}\,\mathrm{d}r \tag{6.62}$$

$$= \int_{|r-r_j|>\delta_N} 2r\,\mathrm{e}^{-M\left((q(r)-2\tau(j)\ln(r))-(q(r_{\tau(j)})-2\tau(j)\ln(r_{\tau(j)}))\right)}\,\mathrm{d}r, \tag{6.63}$$

$$= 2\left(\mathrm{e}^{-q(r_{\tau(j)})}\,r_{\tau(j)}^{-2\tau(j)}\right)^M \int_{|r-r_j|>\delta_N} \left(\mathrm{e}^{-q(r)}\,r^{2\tau(j)}\right)^M r\mathrm{d}r. \tag{6.64}$$

By assumption, we chose M to be a big enough number such that the integral above converges. If that is true,

$$(1) = \int_{|r-r_j|>\delta_N} 2r\,\mathrm{e}^{-NV_{\tau(j)}(r)}\,\mathrm{d}r \leq \mathrm{e}^{-NV_{\tau(j)}(r_{\tau(j)})}\,\epsilon_N, \tag{6.65}$$

with $\epsilon_N = \mathcal{O}(\mathrm{e}^{-c(\ln(N))^2})$. □

We now move on to the value-contributing part of the integral. The other order term should be absolved in this one if Theorem 2 is to be right.

Lemma 4. *For the second part of the integral we have, for* (2) *as in Equation 6.41 and* $0 \leq j \leq N-1$*, asymptotically on* N*,*

$$(2) := \int_{|r-r_j|<\delta_N} 2r\,\mathrm{e}^{-NV_{\tau(j)}(r)}\,\mathrm{d}r = \frac{\mathrm{e}^{-NV_{\tau(j)}(r_{\tau(j)})}}{\sqrt{N}} \left(\frac{8\pi r_{\tau(j)}^2}{V_{\tau(j)}^{(2)}(r_{\tau(j)})}\right)^{\frac{1}{2}} \left[1 + \frac{1}{N}\mathfrak{B}(r_{\tau(j)}) + \mathcal{O}(N^{-2})\right] \tag{6.66}$$

where $\mathfrak{B}(r_{\tau(j)})$ *is as in Theorem 2.*

Proof. As we're dealing with a symmetric (in relation to the critical point) integral, we might as well set a centralized variable $x = r - r_\tau$. With this, the integral becomes

$$(2) = \int_{-\delta_N}^{\delta_N} 2(x + r_{\tau(j)})\,\mathrm{e}^{-NV_{\tau(j)}(x+r_{\tau(j)})}\,\mathrm{d}x, \tag{6.67}$$

$$= \mathrm{e}^{-NV_{\tau(j)}(r_{\tau(j)})} \int_{-\delta_N}^{\delta_N} 2(x + r_{\tau(j)})\,\mathrm{e}^{-N\left[V_{\tau(j)}(x+r_{\tau(j)})-V_{\tau(j)}(r_{\tau(j)})\right]}\,\mathrm{d}x. \tag{6.68}$$

There is also a question of scale. Our interval grows with $\sqrt{N}/\ln(N)$. By re-scaling the reference variable using $y = \sqrt{N}x$ we get

$$(2) = \frac{\mathrm{e}^{-NV_{\tau(j)}(r_{\tau(j)})}}{\sqrt{N}} \int_{-\ln(N)}^{\ln(N)} 2\left(\frac{y}{\sqrt{N}} + r_{\tau(j)}\right) \mathrm{e}^{-N\left[V_{\tau(j)}\left(\frac{y}{\sqrt{N}}+r_{\tau(j)}\right)-V_{\tau(j)}(r_{\tau(j)})\right]}\,\mathrm{d}y. \tag{6.69}$$

Now, the idea here is quite direct: to use the Taylor expansion for $V_{\tau(j)}\left(\frac{y}{\sqrt{N}} + r_{\tau(j)}\right)$ near the critical point $r_{\tau(j)}$. That is,

$$(2) = \frac{\mathrm{e}^{-NV_{\tau(j)}(r_{\tau(j)})}}{\sqrt{N}} \int_{-\ln(N)}^{\ln(N)} 2\left(\frac{y}{\sqrt{N}} + r_{\tau(j)}\right) \mathrm{e}^{-N\left(\frac{y^2}{2N}V^{(2)}_{\tau(j)}(r_{\tau(j)})+\frac{y^3}{3!N^{\frac{3}{2}}}V^{(3)}_{\tau(j)}(r_{\tau(j)})+\frac{y^4}{4!N^2}V^{(4)}_{\tau(j)}(r_{\tau(j)})+O\left(\left(\frac{y}{\sqrt{N}}\right)^5\right)\right)}\,\mathrm{d}y, \tag{6.70}$$

$$= \frac{\mathrm{e}^{-NV_{\tau(j)}(r_{\tau(j)})}}{(2r_{\tau(j)})^{-1}\sqrt{N}} \int_{-\ln(N)}^{\ln(N)} \left(\frac{y}{r_{\tau(j)}\sqrt{N}} + 1\right) \mathrm{e}^{-\frac{y^2}{2}V^{(2)}_{\tau(j)}(r_{\tau(j)})-\frac{y^3}{3!\sqrt{N}}V^{(3)}_{\tau(j)}(r_{\tau(j)})-\frac{y^4}{4!N}V^{(4)}_{\tau(j)}(r_{\tau(j)})+O\left(\frac{y^5}{N^{\frac{3}{2}}}\right)}\,\mathrm{d}y, \tag{6.71}$$

$$= \frac{\mathrm{e}^{-NV_{\tau(j)}(r_{\tau(j)})}}{(2r_{\tau(j)})^{-1}\sqrt{N}} \int_{-\ln(N)}^{\ln(N)} \left(\frac{y}{r_{\tau(j)}\sqrt{N}} + 1\right) \mathrm{e}^{-\frac{y^2}{2}V^{(2)}_{\tau(j)}(r_{\tau(j)})}\left(\mathrm{e}^{-\frac{y^3}{3!\sqrt{N}}V^{(3)}_{\tau(j)}(r_{\tau(j)})-\frac{y^4}{4!N}V^{(4)}_{\tau(j)}(r_{\tau(j)})+O\left(\frac{y^5}{N^{\frac{3}{2}}}\right)}\right)\mathrm{d}y. \tag{6.72}$$

Now we take the Taylor Series for

$$\begin{aligned} \mathrm{e}^{-\frac{y^3}{3!\sqrt{N}}V^{(3)}_{\tau(j)}(r_{\tau(j)})-\frac{y^4}{4!N}V^{(4)}_{\tau(j)}(r_{\tau(j)})+O\left(\frac{y^5}{N^{\frac{3}{2}}}\right)} &= 1 - \left[\frac{y^3}{3!\sqrt{N}}V^{(3)}_{\tau(j)}(r_{\tau(j)}) + \frac{y^4}{4!N}V^{(4)}_{\tau(j)}(r_{\tau(j)}) + O\left(\frac{y^5}{N^{\frac{3}{2}}}\right)\right] \\ -\frac{1}{2}\left[\frac{y^6}{N}\left(\frac{V^{(3)}_{\tau(j)}(r_{\tau(j)})}{6}\right)^2 + \frac{y^8}{N^2}\left(\frac{V^{(4)}_{\tau(j)}(r_{\tau(j)})}{24}\right)^2 \right.&\left.+ \frac{y^7}{N^3}\left(\frac{V^{(4)}_{\tau(j)}(r_{\tau(j)})}{6}\frac{V^{(3)}_{\tau(j)}(r_{\tau(j)})}{24}\right) + O\left(\frac{y^8}{N^2}\right)\right]. \end{aligned} \tag{6.73}$$

To get the asymptotic expansion (since odd terms vanish) we reorganize the terms in the integral again to obtain

$$\begin{aligned} (2) = 2r_{\tau(j)}\frac{\mathrm{e}^{-NV_{\tau(j)}(r_{\tau(j)})}}{\sqrt{N}} \int_{-\ln(N)}^{\ln(N)} \mathrm{e}^{-\frac{y^2}{2}V^{(2)}_{\tau(j)}(r_{\tau(j)})}\Bigg[1 &- \frac{y^4}{3!r_{\tau(j)}N}V^{(3)}_{\tau(j)}(r_{\tau(j)}) \\ &-\frac{y^4}{4!N}V^{(4)}_{\tau(j)}(r_{\tau(j)}) - \frac{y^6}{2\cdot(3!)^2N}V^{(3)}_{\tau(j)}(r_{\tau(j)})^2 + O\left(\frac{y^6}{N^2}\right)\Bigg]\mathrm{d}y, \end{aligned} \tag{6.74}$$

then, distributing the integral,

$$\begin{aligned} (2) = 2r_{\tau(j)}\frac{\mathrm{e}^{-NV_{\tau(j)}(r_{\tau(j)})}}{\sqrt{N}} &\left[\left(-\frac{V^{(4)}_{\tau(j)}(r_{\tau(j)})}{24N} - \frac{V^{(3)}_{\tau(j)}(r_{\tau(j)})}{6r_{\tau(j)}N}\right)\int_{-\ln(N)}^{\ln(N)} \mathrm{e}^{-\frac{y^2}{2}V^{(2)}_{\tau(j)}(r_{\tau(j)})}y^4 \right. \\ &\left. -\frac{V^{(3)}_{\tau(j)}(r_{\tau(j)})^2}{72N}\int_{-\ln(N)}^{\ln(N)} \mathrm{e}^{-\frac{y^2}{2}V^{(2)}_{\tau(j)}(r_{\tau(j)})}y^6 + \int_{-\ln(N)}^{\ln(N)} \mathrm{e}^{-\frac{y^2}{2}V^{(2)}_{\tau(j)}(r_{\tau(j)})} + O\left(N^{-2}\right)\right]. \end{aligned} \tag{6.75}$$

At this point, the result can be directly evaluated using the Gaussian integrals in Proposition 1. So that we can finally substitute the values to get

$$\begin{aligned}(2) = \sqrt{2\pi} r_{\tau(j)} \frac{\mathrm{e}^{-NV_{\tau(j)}(r_{\tau(j)})}}{\sqrt{N}} & \left[\left(-\frac{V^{(4)}_{\tau(j)}(r_{\tau(j)})}{24N} - \frac{V^{(3)}_{\tau(j)}(r_{\tau(j)})}{6r_{\tau(j)}N} \right) \frac{3}{16} \left(\frac{4}{V^{(2)}_{\tau(j)}(r_{\tau(j)})} \right)^{\frac{5}{2}} \right. \\ & \left. - \frac{V^{(3)}_{\tau(j)}(r_{\tau(j)})^2}{72N} \frac{15}{64} \left(\frac{4}{V^{(2)}_{\tau(j)}(r_{\tau(j)})} \right)^{\frac{7}{2}} + \left(\frac{4}{V^{(2)}_{\tau(j)}(r_{\tau(j)})} \right)^{\frac{1}{2}} + O\left(N^{-2}\right) \right], \end{aligned} \tag{6.76}$$

and, rearranging the terms we should get our result

$$(2) = \frac{\mathrm{e}^{-NV_{\tau(j)}(r_{\tau(j)})}}{\sqrt{N}} \left(\frac{8\pi r^2_{\tau(j)}}{V^{(2)}_{\tau(j)}(r_{\tau(j)})} \right)^{\frac{1}{2}} \left[1 + \frac{1}{N}\mathfrak{B}(r_{\tau(j)}) + O(N^{-2}) \right] \tag{6.77}$$

where we defined $\mathfrak{B}(r_{\tau(j)})$ as

$$\mathfrak{B}(r_{\tau(j)}) = -\frac{V^{(4)}_{\tau(j)}(r_{\tau(j)})}{V^{(2)}_{\tau(j)}(r_{\tau(j)})^2}\frac{1}{8} - \frac{V^{(3)}_{\tau(j)}(r_{\tau(j)})}{V^{(2)}_{\tau(j)}(r_{\tau(j)})^2}\frac{1}{2r_{\tau(j)}} - \frac{V^{(3)}_{\tau(j)}(r_{\tau(j)})^2}{V^{(2)}_{\tau(j)}(r_{\tau(j)})^3}\frac{5}{24}. \tag{6.78}$$

Notice that the error term of Lemma 3 is absorbed into the error order of Equation 6.77. Therefore we have the asymptotic expression for h_j given by Equation 6.39 and the proof for Theorem 2 follows directly. □

6.2.1.2 Asymptotics on the Summation

We return to Equation 6.23 and turn to the calculation of each term of $\sum_{j=0}^{N-1} \ln h_j$ using the logarithm of the expansion calculated previously in Theorem 2

$$\sum_{j=0}^{N-1} -NV_{\tau(j)}(r_{\tau(j)}) + \frac{1}{2}\left(\ln(8\pi r^2_{\tau(j)}) - \ln N - \ln V^{(2)}_{\tau(j)}(r_{\tau(j)}) \right) + \frac{1}{N}\mathfrak{B}(r_{\tau(j)}) + O(N^{-2}). \tag{6.79}$$

That is, we need to prove the following theorem.

Theorem 3. *Asymptotically on N,*

$$\sum_{j=0}^{N-1} \ln h_j = -N^2 I_Q[\mu_Q] - \frac{N}{2}\ln N + N\left(\frac{\ln(8\pi)}{2} - \frac{1}{2}E_Q[\mu_Q] \right) + F_V[A_{[r_0,r_1]}] + O(N^{-1}) \tag{6.80}$$

where $F_V[A_{[r_0,r_1]}]$, $I_Q[\mu_Q]$, $E_Q[\mu_Q]$ are as in Theorem 1.

The proof comes from gathering the terms in Equation 6.79 with its respective expressions derived in Lemmas 5 and 6. We go term by term. Lemma 5 refers to the asymptotic of

$$\sum_{j=0}^{N-1} -NV_{\tau(j)}(r_{\tau(j)}), \tag{6.81}$$

and Lemma 6 refers to the asymptotics of both terms

$$\sum_{j=0}^{N-1}\left(\ln(8\pi r_{\tau(j)}^2)-\ln V^{(2)}_{\tau(j)}(r_{\tau(j)})\right). \tag{6.82}$$

Then, the final expression becomes a question of balancing the results.

Lemma 5. *Asymptotically on N,*

$$-N\sum_{j=0}^{N-1}V_{\tau(j)}(r_{\tau(j)})=-N^2I_Q[\mu_Q]+NU_{\mu_Q}(0)-\frac{1}{6}\ln\left(\frac{r_0}{r_1}\right)+O(N^{-2}) \tag{6.83}$$

where we have set $I_Q[\mu_Q]$ as in Theorem 1 and

$$-2U_{\mu_Q}(0):=\int_{r_0}^{r_1}\ln(s)(sq'(s))'\mathrm{d}s=q(r_1)-q(r_0)+2\ln(r_1). \tag{6.84}$$

Proof. Using the Euler-Maclaurin formula

$$\sum_{j=0}^{N-1}V_{\tau(j)}(r_{\tau(j)})=\int_0^N V_{\tau(t)}(r_{\tau(t)})\mathrm{d}t-\frac{1}{2}\left(V_{\tau(N)}(r_{\tau(N)})-V_{\tau(0)}(r_{\tau(0)})\right)+$$
$$\frac{1}{12}\left[\partial_t V_{\tau(t)}(r_{\tau(t)})|_{t=N}-\partial_t V_{\tau(t)}(r_{\tau(t)})|_{t=0}\right]+O(N^{-3}). \tag{6.85}$$

The first term in Equation 6.85 is solved doing a integral by parts. We remember the expression for the derivative of the potential $V'_\tau(r)=q'(r)-2\tau/r$. Note that in the critical point $r_{\tau(j)}$, we should have $r_{\tau(j)}q'(r_{\tau(j)})=2\tau(j)$. This implies that, for $r(\tau):=r_\tau$, the differential can be written as

$$2\mathrm{d}\tau(j)=\mathrm{d}(r_\tau q'(r_\tau)), \tag{6.86}$$
$$=\mathrm{d}r_\tau q'(r_\tau)+\mathrm{d}r_\tau r_\tau)q''(r_\tau), \tag{6.87}$$
$$=\mathrm{d}r_\tau(q'(r_\tau)+r_\tau q''(r_\tau)). \tag{6.88}$$

And, therefore,

$$\frac{\mathrm{d}r(\tau)}{\mathrm{d}\tau}=\frac{2}{r(\tau)V^{(2)}_\tau(r(\tau))}=\frac{2}{(rq'(r))'}\bigg|_{r=r(\tau)}. \tag{6.89}$$

We then are able to rewrite, setting $s=r_{\tau(t)}$,

$$\int_0^N V_{\tau(t)}(r_{\tau(t)})\mathrm{d}t=\int_0^N(q(r_{\tau(t)})-2\tau(t)\ln(r_{\tau(t)}))\mathrm{d}t, \tag{6.90}$$
$$=\frac{1}{2}\int_{r_0}^{r_1}(q(s)-sq'(s)\ln(s))sV^{(2)}(s)N\mathrm{d}s, \tag{6.91}$$
$$=N\left[\frac{1}{2}\int_{r_0}^{r_1}q(s)sV^{(2)}(s)\mathrm{d}s-\frac{1}{2}\int_{r_0}^{r_1}s^2q'(s)\ln(s)V^{(2)}(s)\mathrm{d}s\right], \tag{6.92}$$

and finally using that $r_1 q'(r_1) = 2$ and $r_0 q'(r_0) = 0$ we can apply the integration by parts on the second term to get

$$\frac{1}{2}\int_{r_0}^{r_1} s^2 q'(s)\ln(s)V^{(2)}(s)\mathrm{d}s = \frac{1}{2}\int_{r_0}^{r_1} s^2 q'(s)\ln(s)\frac{1}{s}(sq'(s))'\mathrm{d}s, \tag{6.93}$$

$$= \frac{1}{2}\int_{r_0}^{r_1} sq'(s)\ln(s)(sq'(s))'\mathrm{d}s, \tag{6.94}$$

$$= \frac{1}{4}\int_{r_0}^{r_1} ((sq'(s))^2)'\ln(s)\mathrm{d}s, \tag{6.95}$$

$$= \frac{1}{4}(sq'(s))^2\ln(s)\Big|_{r_0}^{r_1} - \frac{1}{4}\int_{r_0}^{r_1} (sq'(s))^2\frac{1}{s}\mathrm{d}s \tag{6.96}$$

$$= \ln(r_1) - \frac{1}{4}\int_{r_0}^{r_1} sq'(s)^2\mathrm{d}s, \tag{6.97}$$

$$= \ln(r_1) - \frac{1}{2}q(r_1) + \frac{1}{4}\int_{r_0}^{r_1} q(s)(sq'(s))'\mathrm{d}s. \tag{6.98}$$

Leaving us with

$$\int_0^N V_{\tau(t)}(r_{\tau(t)})\mathrm{d}t = N\left[\frac{1}{2}\int_{r_0}^{r_1} q(s)sV^{(2)}(s)\mathrm{d}s - \ln(r_1) + \frac{1}{2}q(r_1) - \frac{1}{4}\int_{r_0}^{r_1} q(s)(sq'(s))'\mathrm{d}s\right], \tag{6.99}$$

$$= N\left[\frac{1}{4}\int_{r_0}^{r_1} q(s)sV^{(2)}(s)\mathrm{d}s - \ln(r_1) + \frac{1}{2}q(r_1)\right], \tag{6.100}$$

$$=: NI_Q[\mu_Q]. \tag{6.101}$$

We also need an expression for $V_{\tau(j)}(r_{\tau(j)}) - V_{\tau(0)}(r_{\tau(0)})$. This is direct noting that

$$V_{\tau(j)}(r_{\tau(j)}) - V_{\tau(0)}(r_{\tau(0)}) = V_1(r_1) - V_0(r_0) = q(r_1) - q(r_0) + 2\ln(r_1) =: 2U_{\mu_Q}(0). \tag{6.102}$$

And finally we can the remaining term , we use the Leibniz rule and obtain

$$\partial_t V_{\tau(t)}(r_{\tau(t)})|_{t=N} = \partial_t\left(q(r_{\tau(t)}) - 2\tau(t)\ln(r_{\tau(t)})\right)_{t=N}, \tag{6.103}$$

$$= \partial_\tau\left(q(r_{\tau(t)}) - 2\tau(t)\ln(r_{\tau(t)})\right)_{\tau=1}\partial_t\left(\tau(t)\right)_{t=N}, \tag{6.104}$$

$$= \frac{1}{N}\left(\partial_\tau(q(r_\tau)) - 2\left(\ln(r_\tau) + \tau\partial_\tau\left(\ln(r_\tau)\right)_{\tau=1}\right)\right), \tag{6.105}$$

$$= \frac{1}{N}\left[\partial_t(q(r_{\tau(t)}) - 2\ln(r_{\tau(t)}))_{\tau=1} - 2\ln(r_1)\right] \tag{6.106}$$

and

$$\partial_t V_{\tau(t)}(r_{\tau(t)})|_{t=0} = \frac{1}{N}\left[\partial_t(q(r_{\tau(t)}) - 2\ln(r_{\tau(t)}))_{\tau=0} - 2\ln(r_0)\right]. \tag{6.107}$$

Since we can calculate, from $r_1 q'(r_1) = 2$ and $r_0 q'(r_0) = 0$

$$\partial_\tau q(r_\tau)|_{\tau=0} = \left(\frac{2q'(r)}{(rq'(r))'}\right)_{r_0} = 0, \tag{6.108}$$

$$\partial_\tau q(r_\tau)|_{\tau=1} = \left(\frac{2q'(r)}{(rq'(r))'}\right)_{r_1} = \frac{4}{r_1(q'(r_1) + r_1 q''(r_1))}, \tag{6.109}$$

$$\partial_\tau(2\ln(r_\tau))|_{\tau=1} = \left(\frac{4}{r(rq'(r))'}\right)_{r_1} = \frac{4}{r_1(q'(r_1)+r_1q''(r_1))}, \tag{6.110}$$

we can also express the difference as

$$\partial_t V_{\tau(t)}(r_{\tau(t)})|_{t=N} - \partial_t V_{\tau(t)}(r_{\tau(t)})|_{t=0} = \frac{1}{12N}\left[2(\ln(r_0)-\ln(r_1))\right] = \frac{1}{6N}\ln\left(\frac{r_0}{r_1}\right). \tag{6.111}$$

The proof for the summation $\sum_{j=0}^{N-1} V_{\tau(j)}(r_{\tau(j)})$ is complete if one gathers the terms in Equation 6.101, Equation 6.111 and Equation 6.102,

□

Lemma 6. *Asymptotically on N,*

$$\sum_{j=0}^{N-1} \ln V^{(2)}_{\tau(j)}(r_{\tau(j)}) = NE_Q[\mu_Q] - \frac{1}{2}\ln\left(\frac{V_1^{(2)}(r_1)}{V_0^{(2)}(r_0)}\right) + O(N^{-1}) \tag{6.112}$$

and

$$\sum_{j=0}^{N-1} \ln(8\pi r^2_{\tau(j)}) = N\ln(8\pi) + 2\left(-NU_{\mu_Q}(0) - \frac{1}{2}\ln\left(\frac{r_1}{r_0}\right) + O(N^{-1})\right) \tag{6.113}$$

where $U_{\mu_Q}(0)$ as in Lemma 5 and $E_Q[\mu_Q]$ as in Theorem 1.

Proof. Lets start now with the second summation, to which we apply Euler-Maclaurin again

$$\sum_{j=0}^{N-1} \ln V^{(2)}_{\tau(j)}(r_{\tau(j)}) = \int_0^N \ln V^{(2)}_{\tau(t)}(r_{\tau(t)})\mathrm{d}t - \frac{1}{2}\ln\left(\frac{V_1^{(2)}(r_1)}{V_0^{(2)}(r_0)}\right) + \frac{1}{12}\left[\partial_t \ln V^{(2)}_{\tau(t)}(r_{\tau(t)})|_{t=N} - \partial_t \ln V^{(2)}_{\tau(t)}(r_{\tau(t)})|_{t=0}\right] + O(N^{-3}). \tag{6.114}$$

We first tackle the integral on the right hand side, changing variables as $s = r_{\tau(t)}$ we would have

$$\int_0^N \ln V^{(2)}_{\tau(t)}(r_{\tau(t)})\mathrm{d}t = \frac{N}{2}\int_{r_0}^{r_1} \ln(V^{(2)}(s))(sq'(s))'\mathrm{d}s. \tag{6.115}$$

And, noticing that

$$\partial_t \ln V^{(2)}_{\tau(t)}(r_{\tau(t)}) = \partial_V\left(\ln V^{(2)}_{\tau(t)}(r_{\tau(t)})\right)\partial_r\left(V^{(2)}_{\tau(t)}(r_{\tau(t)})\right)\partial_\tau\left(r_{\tau(t)}\right) = \frac{\left(\partial_r V^{(2)}\right)(r_{\tau(t)})}{r_{\tau(t)}\left(V^{(2)}_{\tau(t)}(r_{\tau(t)})\right)^2}, \tag{6.116}$$

we have for the third term on the right hand side,

$$\partial_t \ln V^{(2)}_{\tau(t)}(r_{\tau(t)})|_{t=N} - \partial_t \ln V^{(2)}_{\tau(t)}(r_{\tau(t)})|_{t=0} = \frac{1}{2N}\left(\frac{\left(\partial_r V^{(2)}\right)(r_1)}{r_1\left(V_1^{(2)}(r_1)\right)^2} - \frac{\left(\partial_r V^{(2)}\right)(r_0)}{r_0\left(V_0^{(2)}(r_0)\right)^2}\right) \tag{6.117}$$

$$= O(N^{-1}). \tag{6.118}$$

Therefore,

$$\sum_{j=0}^{N-1} \ln V^{(2)}_{\tau(j)}(r_{\tau(j)}) = \frac{N}{2}\int_{r_0}^{r_1} \ln(V^{(2)}(s))(sq'(s))'\mathrm{d}s - \frac{1}{2}\ln\left(\frac{V_1^{(2)}(r_1)}{V_0^{(2)}(r_0)}\right) + O(N^{-1}). \tag{6.119}$$

Similarly, for the first summation, we apply Euler-Maclaurin again

$$\sum_{j=0}^{N-1} \ln(8\pi r^2_{\tau(j)}) = \ln(8\pi) + 2\Bigg(\int_0^N \ln(r_{\tau(t)})\mathrm{d}t - \frac{1}{2}\ln\left(\frac{r_1}{r_0}\right) + \frac{1}{12}\left[\partial_t \ln r_{\tau(t)}|_{\tau=1} - \partial_t \ln r_{\tau(t)}|_{\tau=0}\right] + O(N^{-3})\Bigg). \tag{6.120}$$

We first tackle the integral on the right hand side, changing variables as $s = r_{\tau(t)}$ we would have

$$\int_0^N \ln r_{\tau(t)}\mathrm{d}t = \frac{N}{2}\int_{r_0}^{r_1} \ln(s)(sq'(s))'\mathrm{d}s. \tag{6.121}$$

And, noticing that

$$\partial_t \ln r_{\tau(t)} = \partial_r\left(\ln(r_{\tau(t)})\right)\partial_\tau\left(r_{\tau(t)}\right) = \frac{2}{V^{(2)}_{\tau(t)}(r_{\tau(t)})\left(r_{\tau(t)}\right)^2}, \tag{6.122}$$

we have for the third term on the right hand side,

$$\partial_t \ln r_{\tau(t)}|_{\tau=1} - \partial_t \ln r_{\tau(t)}|_{\tau=0} = \frac{1}{2N}\left(\frac{2}{V^{(2)}(r_1)\,(r_1)^2} - \frac{2}{V^{(2)}(r_0)\,(r_0)^2}\right) = O(N^{-1}). \tag{6.123}$$

Then

$$\sum_{j=0}^{N-1} \ln(8\pi r^2_{\tau(j)}) = N\ln(8\pi) + 2\left(\frac{N}{2}\int_{r_0}^{r_1} \ln(s)(sq'(s))'\mathrm{d}s - \frac{1}{2}\ln\left(\frac{r_1}{r_0}\right) + O(N^{-1})\right). \tag{6.124}$$

□

Proof. (Theorem 3) We only need to notice now that

$$\frac{1}{N}\sum_{j=0}^{N-1} \mathfrak{B}(r_{\tau(j)}) = \frac{1}{4}\int_0^{r_1} \mathfrak{B}(s)(sq'(s))'\mathrm{d}s + O(N^{-2}) \tag{6.125}$$

$$= \frac{1}{4}\int_0^{r_1} \mathfrak{B}(s)sV^{(2)}(s)\mathrm{d}s + O(N^{-2}). \tag{6.126}$$

Using the definition for $\mathfrak{B}(r_{\tau(j)})$ and integrating by parts,

$$\frac{1}{N}\sum_{j=0}^{N-1} \mathfrak{B}(r_{\tau(j)}) = \frac{1}{4}\int_{r_0}^{r_1}\left[-\frac{V^{(4)}(r)}{V^{(2)}(r)^2}\frac{1}{8} - \frac{V^{(3)}(r)}{V^{(2)}(r)^2}\frac{1}{2r} - \frac{V^{(3)}(r)^2}{V^{(2)}(r)^3}\frac{5}{24}\right] rV^{(2)}(r)\mathrm{d}r + O(N^{-2}), \tag{6.127}$$

$$= -\frac{1}{32}\int_{r_0}^{r_1} \frac{V^{(4)}(r)}{V^{(2)}(r)} r\mathrm{d}r - \frac{1}{8}\int_{r_0}^{r_1} \frac{V^{(3)}(r)}{V^{(2)}(r)}\mathrm{d}r - \frac{5}{96}\int_{r_0}^{r_1} \frac{V^{(3)}(r)^2}{V^{(2)}(r)^2} r\mathrm{d}r + O(N^{-2}), \tag{6.128}$$

$$= -\frac{1}{8}\ln\left(\frac{V_1^{(2)}(r_1)}{V_0^{(2)}(r_0)}\right) - \frac{1}{32}\int_{r_0}^{r_1} \frac{V^{(4)}(r)}{V^{(2)}(r)} r\mathrm{d}r - \frac{5}{96}\int_{r_0}^{r_1} \frac{V^{(3)}(r)^2}{V^{(2)}(r)^2} r\mathrm{d}r + O(N^{-2}), \tag{6.129}$$

where

$$\int_{r_0}^{r_1}\frac{V^{(4)}(r)}{V^{(2)}(r)}r\mathrm{d}r=\frac{r_1V_1^{(3)}(r_1)}{V_1^{(2)}(r_1)}-\frac{r_0V_0^{(3)}(r_0)}{V_0^{(2)}(r_0)}-\ln\left(\frac{V_1^{(2)}(r_1)}{V_0^{(2)}(r_0)}\right)+\int_{r_0}^{r_1}\frac{V^{(3)}(r)^2}{V^{(2)}(r)^2}r\mathrm{d}r. \tag{6.130}$$

Therefore

$$\frac{1}{N}\sum_{j=0}^{N-1}\mathfrak{B}(r_{\tau(j)})=-\frac{3}{32}\ln\left(\frac{V_1^{(2)}(r_1)}{V_0^{(2)}(r_0)}\right)-\frac{1}{32}\left(\frac{r_1V_1^{(3)}(r_1)}{V_1^{(2)}(r_1)}-\frac{r_0V_0^{(3)}(r_0)}{V_0^{(2)}(r_0)}\right)-\frac{1}{12}\int_{r_0}^{r_1}\frac{V^{(3)}(r)^2}{V^{(2)}(r)^2}r\mathrm{d}r+O(N^{-2}), \tag{6.131}$$

and the theorem follows from

$$\begin{aligned}\sum_{j=0}^{N-1}\ln h_j=&-N^2I_Q[\mu_Q]+NU_{\mu_Q}(0)-\frac{1}{6}\ln\left(\frac{r_0}{r_1}\right)-\frac{N}{2}\ln(N)-\frac{1}{2}\left(NE_Q[\mu_Q]-\frac{1}{2}\ln\left(\frac{V_1^{(2)}(r_1)}{V_0^{(2)}(r_0)}\right)\right)\\&+\frac{1}{2}\left(N\ln(8\pi)+2\left(-NU_{\mu_Q}(0)-\frac{1}{2}\ln\left(\frac{r_1}{r_0}\right)\right)\right)-\frac{3}{32}\ln\left(\frac{V_1^{(2)}(r_1)}{V_0^{(2)}(r_0)}\right)\\&-\frac{1}{32}\left(\frac{r_1V_1^{(3)}(r_1)}{V_1^{(2)}(r_1)}-\frac{r_0V_0^{(3)}(r_0)}{V_0^{(2)}(r_0)}\right)-\frac{1}{12}\int_{r_0}^{r_1}\frac{V^{(3)}(r)^2}{V^{(2)}(r)^2}r\mathrm{d}r+O(N^{-1})\end{aligned} \tag{6.132}$$

that reduces to

$$\begin{aligned}\sum_{j=0}^{N-1}\ln h_j=&-N^2I_Q[\mu_Q]-\frac{N}{2}\ln(N)+N\left(\frac{\ln(8\pi)}{2}-\frac{1}{2}E_Q[\mu_Q]\right)-\frac{2}{3}\ln\left(\frac{r_0}{r_1}\right)+\frac{5}{32}\ln\left(\frac{V_1^{(2)}(r_1)}{V_0^{(2)}(r_0)}\right)\\&-\frac{1}{32}\left(\frac{r_1V_1^{(3)}(r_1)}{V_1^{(2)}(r_1)}-\frac{r_0V_0^{(3)}(r_0)}{V_0^{(2)}(r_0)}\right)-\frac{1}{12}\int_{r_0}^{r_1}\frac{V^{(3)}(r)^2}{V^{(2)}(r)^2}r\mathrm{d}r+O(N^{-1}).\end{aligned} \tag{6.133}$$

□

6.2.1.3 Free Energy Annulus

We can finally prove Theorem 1.

Proof. (Theorem 1) From Theorem 2,

$$\begin{aligned}\sum_{j=0}^{N-1}\ln(h_j)=&-N^2I_Q[\mu_Q]-\frac{N}{2}\ln(N)+N\left(\frac{\ln(8\pi)}{2}-\frac{1}{2}E_Q[\mu_Q]\right)-\frac{2}{3}\ln\left(\frac{r_0}{r_1}\right)+\frac{5}{32}\ln\left(\frac{V_1^{(2)}(r_1)}{V_0^{(2)}(r_0)}\right)\\&-\frac{1}{32}\left(\frac{r_1V_1^{(3)}(r_1)}{V_1^{(2)}(r_1)}-\frac{r_0V_0^{(3)}(r_0)}{V_0^{(2)}(r_0)}\right)-\frac{1}{12}\int_{r_0}^{r_1}\frac{V^{(3)}(r)^2}{V^{(2)}(r)^2}r\mathrm{d}r+O(N^{-1}).\end{aligned} \tag{6.134}$$

We can get the asymptotic expansion for the Free energy using that

$$\ln(N!)=N\ln(N)-N+\frac{1}{2}\ln(N)+\frac{1}{2}\ln(2\pi)+O(N^{-1}) \tag{6.135}$$

that is,

$$\ln \mathcal{Z}_N = \ln N! + \sum_{j=0}^{N-1} \ln h_j \tag{6.136}$$

$$= N\ln(N) - N + \frac{1}{2}\ln(N) + \frac{1}{2}\ln(2\pi) + \left(-N^2 I_Q[\mu_Q] - \frac{N}{2}\ln(N) + N\left(\frac{\ln(8\pi)}{2} - \frac{1}{2}E_Q[\mu_Q]\right)\right.$$
$$\left. - \frac{2}{3}\ln\left(\frac{r_0}{r_1}\right) + \frac{5}{32}\ln\left(\frac{V_1^{(2)}(r_1)}{V_0^{(2)}(r_0)}\right) - \frac{1}{32}\left(\frac{r_1 V_1^{(3)}(r_1)}{V_1^{(2)}(r_1)} - \frac{r_0 V_0^{(3)}(r_0)}{V_0^{(2)}(r_0)}\right) - \frac{1}{12}\int_{r_0}^{r_1} \frac{V^{(3)}(r)^2}{V^{(2)}(r)^2} r\mathrm{d}r\right) + O(N^{-1}) \tag{6.137}$$

$$= -N^2 I_Q[\mu_Q] + \frac{N}{2}\left(\ln(8\pi) - E_Q[\mu_Q] - 2\right) + \frac{N}{2}\ln(N) + \frac{1}{2}\ln(N) + \frac{\ln(2\pi)}{2}$$
$$- \frac{2}{3}\ln\left(\frac{r_0}{r_1}\right) + \frac{5}{32}\ln\left(\frac{V_1^{(2)}(r_1)}{V_0^{(2)}(r_0)}\right) - \frac{1}{32}\left(\frac{r_1 V_1^{(3)}(r_1)}{V_1^{(2)}(r_1)} - \frac{r_0 V_0^{(3)}(r_0)}{V_0^{(2)}(r_0)}\right) - \frac{1}{12}\int_{r_0}^{r_1} \frac{V^{(3)}(r)^2}{V^{(2)}(r)^2} r\mathrm{d}r O(N^{-1}) \tag{6.138}$$

and the theorem follows. □

6.2.2 Asymptotics - Disk

In this case, we consider $r_0 = 0$. As before, the measure in Equation 6.2 to be the law for distribution of the particles in this model. When we say $r_0 = 0$, we mean that the support for this configuration space of particles is a disk in the complex plane. The expression we get is different from taking the limit in the case $r_0 > 0$. (BYUN; KANG; SEO, 2023)

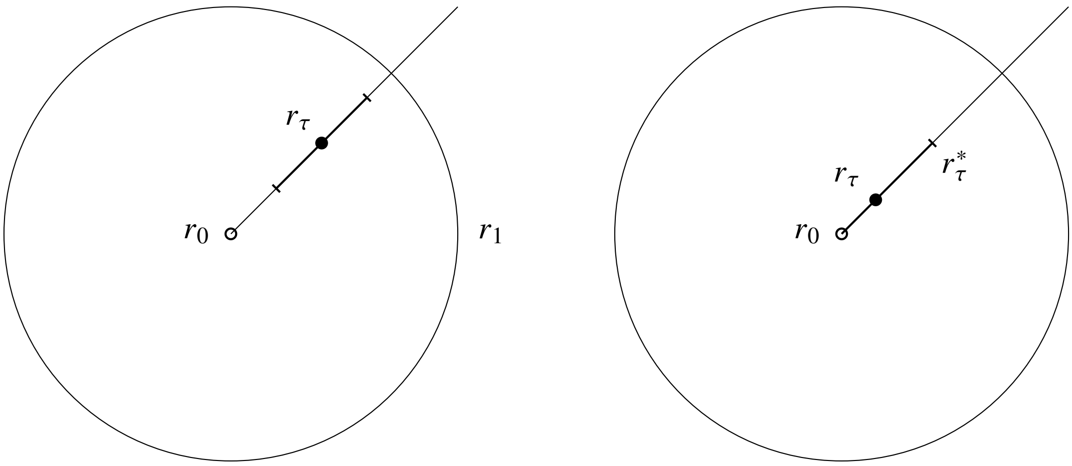

Figure 2 – The figure depicts an disk referring to the equilibrium measure when $r_0 = 0$. As the complex plane integral is reduced to a line integral, we draw the domain change and the point of interest r_τ where V has a critical point. Around this point we depict the region around the minima, which in the method that should account for the main order of the integral we solve for. The left image corresponds to the case when $r_{j/N} >> N^{-\epsilon}$ and the right image corresponds to $r_{j/N} << N^{-\epsilon}$. In this case $r*_\tau$ will define the contributing region to the integral.

Theorem 4. *For the annulus case, where $r_0 = 0$, we should have the following, asymptotically on N, for the free energy*

$$\ln \mathcal{Z}_N = \tag{6.139}$$

where we have set

$$I_Q[\mu_Q] := \left[\frac{1}{4}\int_{r_0}^{r_1} q(s)sV^{(2)}(s)\mathrm{d}s - \ln(r_1) + \frac{1}{2}q(r_1)\right], \tag{6.140}$$

$$E_Q[\mu_Q] := \frac{1}{2}\int_{r_0}^{r_1} \ln(V^{(2)}(s))(sq'(s))'\mathrm{d}s \tag{6.141}$$

and

$$F_V[D_{r_1}] = \frac{1}{6}\ln\left(\frac{4}{V_0^{(2)}(0)}\right) - \frac{1}{3}\ln(r_1) + \frac{5}{32}\ln\left(\frac{V_1^{(2)}(r_1)}{V_0^{(2)}(r_0)}\right) - \frac{1}{32}\frac{r_1 V_1^{(3)}(r_1)}{V_1^{(2)}(r_1)} - \frac{1}{12}\int_{r_0}^{r_1}\frac{V^{(3)}(r)^2}{V^{(2)}(r)^2} r\mathrm{d}r. \tag{6.142}$$

Again, we start by studying the asymptotic for h_j and then for the summation $\sum \ln(h_j)$. This theorem then should follow from the following lemmas concerning this two asymptotics.

6.2.2.1 Asymptotic for the norming constant

In the asymptotic expansions of h_j in this regimen, one should distinguish the following two cases depending on a small constant $1/5 > \epsilon > 0$.

- Case 1: $r_{j/N} >> N^{-\epsilon}$. For the annular droplet case where $r_0 > 0$, this case covers all $j = 0, 1, \cdots, N-1$. On the other hand, for the disk droplet case where $r_0 = 0$, this case covers only $j = m_N, m_N + 1, \cdots, N-1$ for $m_N = N^\epsilon$ with some $\epsilon > 0$;
- Case 2: $r_{j/N} << N^{-\epsilon}$. This covers the remaining disk droplet case with $j = 0, 1, \cdots, m_N - 1$. Notably, the asymptotic expansion involves gamma functions in this case.

We separate this in two proofs, the first one for $j = m_N, m_N + 1, \cdots, N-1$. One might assume, and would be mainly correct, that nothing is that different in the disk and annulus regimes if the points are distant from r_0. The following results are direct adaptations of the previous arguments.

Theorem 5. *For $N - 1 \geq j \geq m_N$, asymptotically on N,*

$$h_j = \frac{\mathrm{e}^{-NV_{\tau(j)}(r_{\tau(j)})}}{\sqrt{N}}\left(\frac{8\pi r_{\tau(j)}^2}{V_{\tau(j)}^{(2)}(r_{\tau(j)})}\right)^{\frac{1}{2}}\left(1 + \frac{1}{N}\mathfrak{B}(r_{\tau(j)}) + \epsilon_{N,1}\right) \tag{6.143}$$

where

$$\mathfrak{B}(r_{\tau(j)}) := -\frac{V^{(4)}_{\tau(j)}(r_{\tau(j)})}{V^{(2)}_{\tau(j)}(r_{\tau(j)})^2}\frac{1}{8} - \frac{V^{(3)}_{\tau(j)}(r_{\tau(j)})}{V^{(2)}_{\tau(j)}(r_{\tau(j)})^2}\frac{1}{2r_{\tau(j)}} - \frac{V^{(3)}_{\tau(j)}(r_{\tau(j)})^2}{V^{(2)}_{\tau(j)}(r_{\tau(j)})^3}\frac{5}{24}, \tag{6.144}$$

and

$$\epsilon_{N,1} := \mathcal{O}(j^{-\frac{3}{2}}(\ln N)^{\alpha}) \tag{6.145}$$

for some $\alpha > 0$.

To actually expand on the details, we need some useful facts. We will make a lot of use of the behavior for r_τ, expressed in Proposition 2.

Proposition 2. *Due to strict subharmonicity of* Q *in a neighborhood of the droplet,* r_τ*, defined as before, satisfies*

$$r_\tau = \left(\frac{2\tau}{q''(0)}\right)^{\frac{1}{2}} + \mathcal{O}(\tau) = \left(\frac{4\tau}{V_0^{(2)}(0)}\right)^{\frac{1}{2}} + \mathcal{O}(\tau) \tag{6.146}$$

or more simply

$$r_\tau = \mathcal{O}(\tau^{\frac{1}{2}}) \tag{6.147}$$

as $\tau \to 0$.

Proof. This is a inverse function theorem with a Taylor expansion for equalizing powers. □

We then start as usual dividing the integral into two parts

$$h_j = \int_0^\infty 2r^{2j+1}\,\mathrm{e}^{-Nq(r)}\,\mathrm{d}r = \int_0^\infty 2r\,\mathrm{e}^{-NV_{\tau(j)}(r)}\,\mathrm{d}r, \tag{6.148}$$

$$= \int_{r_{\tau(j)}-\delta_N}^{r_{\tau(j)}+\delta_N} 2r\,\mathrm{e}^{-NV_{\tau(j)}(r)}\,\mathrm{d}r + \int_{|r-r_{\tau(j)}|>\delta_N} 2r\,\mathrm{e}^{-NV_{\tau(j)}(r)}\,\mathrm{d}r =: (1)+(2). \tag{6.149}$$

Let's start with the outer integral, which we should be able to control the growth.

Lemma 7. *Consider the case* $m_N \le j \le N-1$. *If we choose* M *to be such that*

$$\int_{|r-r_j|>\delta_N} \left(\mathrm{e}^{-q(r)}r^2\right)^M r\mathrm{d}r < \infty, \tag{6.150}$$

then, with (2) *as in Equation 6.149, for the error-bound integral we have the asymptotic for* N *given by*

$$(2) := \int_{|r-r_{\tau(j)}|>\delta_N} 2r\,\mathrm{e}^{-NV_{\tau(j)}(r)}\,\mathrm{d}r = \mathrm{e}^{-NV_{\tau(j)}(r_{\tau(j)})}\,\mathcal{O}\left(\mathrm{e}^{c_2(\ln(N))^2}\right). \tag{6.151}$$

for some positive constant c_2.

Proof. Note that for $d \geq 1$, we can choose C_1 a constant uniformly for all τ and write

$$|V_\tau^{(d)}(r_\tau)| = |q^{(d)}(r_\tau) + (-1)^d 2\tau(d-1)!\, r_\tau^{-d}| \leq C_1(1+\tau^{-\frac{d}{2}+1}). \tag{6.152}$$

Firstly, for $m_N \leq j < N$, we have

$$r_{\tau(j+1/2)} - r_{\tau(j)} = O((jN)^{-\frac{1}{2}}). \tag{6.153}$$

Since the potential is increasing in (r_τ, ∞), the Taylor series expansion for V_τ is for all $r > r_{\tau(j)} + \delta_N$

$$V_{\tau(j+1)}(r) \geq V_{\tau(j+1)}(r_{\tau(j)} + \delta_N), \tag{6.154}$$

$$= V_{\tau(j+1)}(r_{\tau(j+1)}) + \frac{1}{2}V^{(2)}_{\tau(j+1)}(r_{\tau(j+1)})(r_{\tau(j)} + \delta_N - r_{\tau(j+1)})^2 + O(\tau(j)^{-1/2}\delta_N^3). \tag{6.155}$$

where $O(\tau(j)^{-1/2}\delta_N^3) = O(N^{-1}j^{-1/2}(\ln(N))^3)$. Similarly, since V_τ is decreasing in $(0, r_\tau)$, we have for all $r < r_{\tau(j)} - \delta_N$

$$V_{\tau(j+1)}(r) \geq V_{\tau(j+1)}(r_{\tau(j)} - \delta_N), \tag{6.156}$$

$$= V_{\tau(j+1)}(r_{\tau(j+1)}) + \frac{1}{2}V^{(2)}_{\tau(j+1)}(r_{\tau(j+1)})(r_{\tau(j)} - \delta_N - r_{\tau(j+1)})^2 + O(\tau(j)^{-1/2}\delta_N^3). \tag{6.157}$$

Using the Taylor series for V_τ again, we have

$$V_{\tau(j+1)}(r_{\tau(j+1)}) - V_{\tau(j)}(r_{\tau(j)}) = V_{\tau(j)}(r_{\tau(j+1)}) - V_{\tau(j)}(r_{\tau(j)}) - \frac{1}{N}\ln r_{\tau(j+1)}, \tag{6.158}$$

$$= O((jN)^{-1}) - \frac{1}{N}\ln r_{\tau(j+1)}. \tag{6.159}$$

Thus, if follows from the last two inequalities that

$$e^{NV_{\tau(j)}(r_{\tau(j)})}\int_{|r-r_{\tau(j)}|>\delta_N} 2r^{2j+1}\, e^{-Nq(r)}\, \mathrm{d}r\mathrm{d}r = \int_{|r-r_{\tau(j)}|>\delta_N} 2\, e^{-N\left(V_{\tau(j+1/2)}(r) - V_{\tau(j)}(r_{\tau(j)})\right)}\, \mathrm{d}r, \tag{6.160}$$

$$\int_{|r-r_{\tau(j)}|>\delta_N} 2\, e^{-(N-M)\left(V_{\tau(j+1/2)}(r) - V_{\tau(j)}(r_{\tau(j)})\right)}\, e^{-M\left(V_{\tau(j+1/2)}(r) - V_{\tau(j)}(r_{\tau(j)})\right)}\, \mathrm{d}r, \tag{6.161}$$

$$\leq e^{-c_1(\ln(N))^2}\, e^{\frac{N-M}{N}\ln(r_{\tau(j+1/2)})}\int_{|r-r_{\tau(j)}|>\delta_N} 2\, e^{-M\left(V_{\tau(j+1/2)}(r) - V_{\tau(j)}(r_{\tau(j)})\right)}\, \mathrm{d}r, \tag{6.162}$$

$$= O(e^{c_2(\ln(N))^2}). \tag{6.163}$$

for some constants $c_1, c_2 > 0$. The details taken here are very similar to the ones in the annulus case. ◻

Lemma 8. *Consider the case $m_N \le j \le N-1$. For the value-contributing integral we have the asymptotic on N given by*

$$(1) := \int_{r_{\tau(j)}-\delta_N}^{r_{\tau(j)}+\delta_N} 2r\,\mathrm{e}^{-NV_{\tau(j)}(r)}\,\mathrm{d}r = \frac{\mathrm{e}^{-NV_{\tau(j)}(r_{\tau(j)})}}{\sqrt{N}}\left(\frac{8\pi r_{\tau(j)}^2}{V_{\tau(j)}^{(2)}(r_{\tau(j)})}\right)^{\frac{1}{2}}\left(1+\frac{1}{N}\mathfrak{B}(r_{\tau(j)})+\epsilon_{N,1}\right) \tag{6.164}$$

where (1) is as in Equation 6.149, $\mathfrak{B}(r_{\tau(j)})$ is as in Theorem 4 and

$$\epsilon_{N,1} := O(j^{-\frac{3}{2}}(\ln N)^{\alpha}) \tag{6.165}$$

for some $\alpha > 0$.

Proof. We now need to examine the integral near the critical point. That is,

$$\int_{r_{\tau(j)}-\delta_N}^{r_{\tau(j)}+\delta_N} 2r\,\mathrm{e}^{-NV_{\tau(j)}(r)}\,\mathrm{d}r \tag{6.166}$$

$$= \mathrm{e}^{-NV_{\tau(j)}(r_{\tau(j)})}\int_{-\delta_N}^{\delta_N} 2(r_{\tau(j)}+t)\,\mathrm{e}^{-N\left(\frac{1}{2}V_{\tau(j)}^{(2)}(r_{\tau(j)})(t)^2+\frac{1}{3!}V_{\tau(j)}^{(3)}(r_{\tau(j)})(t)^3+\frac{1}{4!}V_{\tau(j)}^{(4)}(r_{\tau(j)})(t)^4+O(\tau(j)^{-\frac{3}{2}}t^5)\right)}\,\mathrm{d}t \tag{6.167}$$

$$= \mathrm{e}^{-NV_{\tau(j)}(r_{\tau(j)})}\int_{-\sqrt{N}\delta_N}^{\sqrt{N}\delta_N} 2\,\mathrm{e}^{-\frac{1}{2}V_{\tau(j)}^{(2)}(r_{\tau(j)})t^2}\left(r_{\tau(j)}+\frac{t}{\sqrt{N}}\right)\left(1-\frac{V_{\tau(j)}^{(3)}(r_{\tau(j)})t^3}{6\sqrt{N}}\right.$$
$$\left.-\frac{V_{\tau(j)}^{(4)}(r_{\tau(j)})t^4}{24N}+\frac{V_{\tau(j)}^{(3)}(r_{\tau(j)})^2t^6}{72N}+\epsilon'_{N,1}\right)\mathrm{d}t \tag{6.168}$$

where $\epsilon_{N,1} = O(j^{-\frac{3}{2}}(\ln N)^{\alpha})$ for some $\alpha > 0$. From here, the expression can be directly identified with Equation 6.74 and the techniques for getting the result are one-to-one with the ones already done in that case. □

Proof. (Theorem 5) This is now only a matter to notice that (2) is absolved in to the error term of (1) and the result follows. □

So far, nothing is considerably different from both cases as we are dealing with point distant from the origin. We must now deal with the more delicate case, for $j \le m_N$. The behavior for h_j, in this regimen, is expressed by the following theorem.

Theorem 6. *For $0 \le j < m_N$, we will have the following asymptotic for the norming constant*

$$h_j = \mathrm{e}^{-Nq(0)}\left[N^{-(j+1)}\Gamma(j+1)\left(\frac{1}{2}q''(0)\right)^{-(j+1)}\left(1+O(N^{-\frac{1}{2}}(j+1/2)^{\frac{3}{2}}(\ln N)^3)\right)+\epsilon_N(j)\right] \tag{6.169}$$

where Γ is the usual Gamma function and

$$\epsilon_N(j) := O\left(\mathrm{e}^{-c_2(j+1)(\ln N)^2}\left(\frac{j+1}{N}(\ln(N))^2\right)^{(j+1)\frac{N-M}{N}}\right) \tag{6.170}$$

with $c_2 > 0$ a constant.

Proof. For $j < m_N$, we need to define some $r_\tau^* := r_\tau \ln(N)$ from which we separate the integral in a important part close to the origin and another part which has a lesser order. Consider then the decomposition

$$h_j = \int_0^\infty 2r^{2j+1}\,\mathrm{e}^{-Nq(r)}\,\mathrm{d}r = \int_0^{r^*_{\tau(j+1)}} 2r^{2j+1}\,\mathrm{e}^{-Nq(r)}\,\mathrm{d}r + \int_{r^*_{\tau(j+1)}}^\infty 2r^{2j+1}\,\mathrm{e}^{-Nq(r)}\,\mathrm{d}r. \tag{6.171}$$

Due to strict subharmonicity of Q in a neighborhood of the droplet, r_τ satisfies $r_\tau = O(\tau^{\frac{1}{2}})$ as $\tau \to 0$, this is given by Proposition 2. Since V_τ has a global minimum at $r = r_\tau$ and increases for larger values, for all $r > r^*_{\tau(j+1)}$ we have

$$V_{\tau(j+1/2)}(r) \geq V_{\tau(j+1)}(r)(r^*_{\tau(j+1)}) = q(r^*_{\tau(j+1)}) - \frac{2j+1}{N}\ln(r^*_{\tau(j+1)}) \tag{6.172}$$

$$= q(0) + \frac{1}{2}q''(0)(r^*_{\tau(j+1)})^2 - \frac{2j+1}{N}\ln(r^*_{\tau(j+1)}) + O(r^*_{\tau(j+1)})^3. \tag{6.173}$$

Note that $q'(0) = 0$ since $\Delta Q(z) \in (0,\infty)$ near the origin and $\Delta Q(z) \propto q'(0)/r + q''(0)$. Take

$$\mathrm{e}^{-Nq(0)}\int_{r^*_{\tau(j+1)}}^\infty 2r^{2j+1}\,\mathrm{e}^{-Nq(r)+Nq(0)}\,\mathrm{d}r = \mathrm{e}^{-Nq(0)}\int_{r^*_{\tau(j+1)}}^\infty 2\,\mathrm{e}^{(2j+1)\ln(r)-Nq(r)+Nq(0)}\,\mathrm{d}r, \tag{6.174}$$

$$= \mathrm{e}^{-Nq(0)}\int_{r^*_{\tau(j+1)}}^\infty 2\,\mathrm{e}^{-N\left(-\frac{2}{N}\left(j+\frac{1}{2}\right)\ln(r)-q(r)+q(0)\right)}\,\mathrm{d}r, \tag{6.175}$$

$$= \int_{r^*_{\tau(j+1)}}^\infty 2\,\mathrm{e}^{-N(V_{\tau(j+1/2)}(r)-q(0))}\,\mathrm{d}r \tag{6.176}$$

and, using the inequality explicited above

$$\mathrm{e}^{-NV_{\tau(j+1)}(r)+Nq(0)} = \mathrm{e}^{-NV_{\tau(j+1)}(r)+Nq(0)+MV_{\tau(j+1)}(r)-MV_{\tau(j+1)}(r)} \tag{6.177}$$

$$= \mathrm{e}^{-(N-M)V_{\tau(j+1)}(r)+Nq(0)-MV_{\tau(j+1)}(r)} \tag{6.178}$$

$$\leq \mathrm{e}^{-(N-M)V_{\tau(j+1)}(r^*)+Nq(0)-MV_{\tau(j+1)}(r)} \tag{6.179}$$

$$= \mathrm{e}^{-(N-M)\left(V_{\tau(j+1)}(r^*)-q(0)\right)}\,\mathrm{e}^{-M\left(V_{\tau(j+1)}(r)-q(0)\right)} \tag{6.180}$$

then

$$\int_{r^*_{\tau(j+1)}}^{\infty} 2\,\mathrm{e}^{-N(V_{\tau(j+1)}(r)-q(0))}\,\mathrm{d}r \tag{6.181}$$

$$\leq \mathrm{e}^{-(N-M)V_{\tau(j+1)}(r^*)}\,\mathrm{e}^{(N-M)q(0)}\int_{r^*_{\tau(j+1)}}^{\infty} 2\,\mathrm{e}^{-MV_{\tau(j+1)}(r)}\,\mathrm{d}r \tag{6.182}$$

$$= \mathrm{e}^{-(N-M)\left[q(r^*_{\tau(j+1)})-\frac{2j+2}{N}\ln(r^*_{\tau(j+1)})\right]}\,\mathrm{e}^{(N-M)q(0)}\int_{r^*_{\tau(j+1)}}^{\infty} 2\,\mathrm{e}^{-MV_{\tau(j+1)}(r)}\,\mathrm{d}r \tag{6.183}$$

$$= \mathrm{e}^{-(N-M)q(r^*_{\tau(j+1)})}\,{r^*_{\tau(j+1)}}^{(2j+2)\frac{N-M}{N}}\,\mathrm{e}^{(N-M)q(0)}\int_{r^*_{\tau(j+1)}}^{\infty} 2\,\mathrm{e}^{-MV_{\tau(j+1)}(r)}\,\mathrm{d}r \tag{6.184}$$

$$= \mathrm{e}^{-(N-M)\left[\frac{q''(0)}{2}(r^*_{\tau(j+1)})^2+O(r^*_{\tau(j+1)})^3\right]}\,{r^*_{\tau(j+1)}}^{(2j+2)\frac{N-M}{N}}\int_{r^*_{\tau(j+1)}}^{\infty} 2\,\mathrm{e}^{-MV_{\tau(j+1)}(r)}\,\mathrm{d}r \tag{6.185}$$

$$\leq \mathrm{e}^{-\frac{(N-M)}{2}q''(0)(r^*_{\tau(j+1)})^2)}\,{r^*_{\tau(j+1)}}^{(2j+2)\frac{N-M}{N}}\int_{r^*_{\tau(j+1)}}^{\infty} 2\,\mathrm{e}^{-MV_{\tau(j+1)}(r)}\,\mathrm{d}r \tag{6.186}$$

$$= \mathrm{e}^{-c_1\frac{(N-M)}{2}r^2_{\tau(j+1)}\ln(N)^2}\,{r^*_{\tau(j+1)}}^{(2j+2)\frac{N-M}{N}}\int_{r^*_{\tau(j+1)}}^{\infty} 2\,\mathrm{e}^{-MV_{\tau(j+1)}(r)}\,\mathrm{d}r := \epsilon_N(j) \tag{6.187}$$

$$\tag{6.188}$$

for some $c_1 > 0$ and M as given in the other proof such that the integral converges. Since $r_\tau = O(\tau^{\frac{1}{2}})$, there is a constant $c_2 > 0$ such that as N goes to infinity

$$\epsilon_N(j) = O\left(\mathrm{e}^{-c_1\frac{(N-M)}{2}\tau(j+1)\ln(N)^2}\,{r^*_{\tau(j+1)}}^{(2j+2)\frac{N-M}{N}}\right) \tag{6.189}$$

$$= O\left(\mathrm{e}^{-c_2(j+1)\ln(N)^2}\,{r^*_{\tau(j+1)}}^{(2j+2)\frac{N-M}{N}}\right) \tag{6.190}$$

$$= O\left(\mathrm{e}^{-c_2(j+1)(\ln N)^2}\left(\frac{j+1}{N}(\ln(N))^2\right)^{(j+1)\frac{N-M}{N}}\right) \tag{6.191}$$

Therefore we obtain

$$h_j = \int_0^\infty 2r^{2j+1}\,\mathrm{e}^{-Nq(r)}\,\mathrm{d}r = \int_0^{r^*_{\tau(j+1)}} 2r^{2j+1}\,\mathrm{e}^{-Nq(r)}\,\mathrm{d} + \mathrm{e}^{-Nq(0)}\,\epsilon_N(j) \tag{6.192}$$

$$= \int_0^{r^*_{\tau(j+1)}} 2r^{2j+1}\,\mathrm{e}^{-N\left(q(0)+\frac{1}{2}q''(0)r^2\right)+O(N(r^*_{\tau(j+1)})^3)}\,\mathrm{d}r + \mathrm{e}^{-Nq(0)}\,\epsilon_N(j) \tag{6.193}$$

$$= \mathrm{e}^{-Nq(0)}\left[\int_0^{r^*_{\tau(j+1)}} 2r^{2j+1}\,\mathrm{e}^{-\frac{1}{2}q''(0)r^2N}\,\mathrm{d}r\left(1+O(N(r^*_{\tau(j+1)})^3)\right)+\epsilon_N(j)\right] \tag{6.194}$$

$$= \mathrm{e}^{-Nq(0)}\left[N^{-(j+1)}\Gamma(j+1)\left(\frac{1}{2}q''(0)\right)^{-(j+1)}\left(1+O(N^{-\frac{1}{2}}(j+1/2)^{\frac{3}{2}}(\ln N)^3)\right)+\epsilon_N(j)\right] \tag{6.195}$$

where Γ is the usual gamma function with integral representation

$$\Gamma(j+1) = \int_0^\infty t^j\,\mathrm{e}^{-t}\,\mathrm{d}t \tag{6.196}$$

and we used $t = 1/2q''(0)r^2N$.

□

6.2.2.2 Asymptotics on the Summation

So now we need to return to Equation 6.23 and calculate each term for $\sum_{j=0}^{N-1} \ln h_j$ using the expansion in terms of Theorem 5 and 6 for the expression

$$\sum_{j=0}^{N} \ln(h_j) = \sum_{j=0}^{m_N-1} \ln(h_j) + \sum_{j=m_N}^{N-1} \ln(h_j). \tag{6.197}$$

Theorem 7. *We have the following asymptotics on N for the summation of $\ln(h_j)$ from 0 to $N-1$,*

$$\sum_{j=0}^{N-1} \ln(h_j) = -N^2 I[\mu_Q] - \frac{1}{2}N\ln(N) - \frac{N}{2}\left(E_Q[\mu_Q] - \ln(8\pi)\right) - \frac{\ln(N)}{12} \\ + \zeta'(-1) + F_V[D_{r_1}] + O(N^{-12}(\ln N)^3). \tag{6.198}$$

where $I_Q[\mu_Q]$, $F_V[D_{r_1}]$ and $E_Q[\mu_Q]$ are as in Theorem 4.

To prove this theorem, we must give the asymptotics for both summations. This means we need both Lemmas 9, referring to the sum from powers 0 to $m_N - 1$, and 10, referring to the sum from m_N to $N-1$. We start by the lower power terms.

Lemma 9. *We have the following asymptotic for the summation of $\ln(h_j)$ from 0 to $m_N - 1$,*

$$\sum_{j=0}^{m_N-1} \ln(h_j) = -m_N N q(0) - \frac{m_N(m_N+1)}{2}\left(\ln(N) + \ln\left(\frac{1}{2}q''(0)\right)\right) + \frac{m_N^2\ln(m_N)}{2} - \frac{3}{4}m_N^2 \\ + \frac{\ln(2\pi)m_N}{2} - \frac{\ln(m_N)}{12} + \zeta'(-1) + O(m_N^{-2} + N^{-\frac{1}{2}(1-5\epsilon)}(\ln(N))^3). \tag{6.199}$$

where ϵ is as in Theorem 6.

Proof. The first term in the right is straightforwardly determined, we need only to write, using Theorem 6,

$$\begin{aligned}\sum_{j=0}^{m_N-1} \ln(h_j) &= -m_N N q(0) - \frac{m_N(m_N+1)}{2}\left(\ln(N) + \ln\left(\frac{1}{2}q^{(2)}(0)\right)\right) \\ &\quad + \ln(G(m_N+1)) + O(N^{-\frac{1}{2}(1-5\epsilon)}(\ln(N))^3) \end{aligned} \tag{6.200}$$

$$\begin{aligned} &= -m_N N q(0) - \frac{m_N(m_N+1)}{2}\left(\ln(N) + \ln\left(\frac{1}{2}q^{(2)}(0)\right)\right) + \frac{m_N^2\ln(m_N)}{2} - \frac{3}{4}m_N^2 \\ &\quad + \frac{\ln(2\pi)m_N}{2} - \frac{\ln(m_N)}{12} + \zeta'(-1) + O(m_N^{-2} + N^{-\frac{1}{2}(1-5\epsilon)}(\ln(N))^3) \end{aligned} \tag{6.201}$$

where G is the Barnes G-function defined recursively as

$$G(z+1) = \Gamma(z)G(z) \tag{6.202}$$

and we have used the asymptotic for the G-Barnes function

$$\ln(G(z+1)) = \frac{z^2\ln(z)}{2} - \frac{3}{4}z^2 + \frac{\ln(2\pi)z}{2} - \frac{\ln(z)}{12} + \zeta'(-1) + O(\frac{1}{z^2}). \tag{6.203}$$

□

Lemma 10. *We have the following asymptotic for the summation of* $\ln(h_j)$ *from* m_N *to* $N-1$,

$$\sum_{j=m_N}^{N-1}\ln(h_j) = -N^2 I[\mu_Q] + \frac{(N-m_N)}{2}\ln\frac{8\pi}{N} + Nm_N q(0) + \frac{3}{4}m_N^2 - \frac{N}{2}E_Q[\mu_Q] - \frac{m_N}{2}(m_N+1)\ln\left(\frac{4}{V_0^{(2)}(0)}\right) - \frac{1}{2}\left(m_N^2 - \frac{1}{6}\right)\ln\left(\frac{m_N}{N}\right) + F_V[D_{r_1}] + O(N^{-12}(\ln N)^3). \tag{6.204}$$

where $I_Q[\mu_Q]$, $F_V[D_{r_1}]$ *and* $E_Q[\mu_Q]$ *are as in Theorem 4.*

The second summation is a bit more detailed, we use the previously defined asymptotics in Theorem 5,

$$\ln(h_j) = -NV_{\tau(j)}(r_{\tau(j)}) + \frac{1}{2}\ln\left(\frac{8\pi r_{\tau(j)}^2}{NV_{\tau(j)}^{(2)}(r_{\tau(j)})}\right) + \frac{1}{N}\mathfrak{B}(r_{\tau(j)}) + O(j^{-\frac{3}{2}}(\ln(N))^{\alpha}) \tag{6.205}$$

and need to determine the asymptotic of three terms in order to get the full summation,

$$-N\sum_{j=m_N}^{N-1} V_{\tau(j)}(r_{\tau(j)}), \quad \sum_{j=m_N}^{N-1}\ln V_{\tau(j)}^{(2)}(r_{\tau(j)}) \quad \text{and} \quad \sum_{j=m_N}^{N-1}\ln(r_{\tau(j)}). \tag{6.206}$$

This is done in Proposition 3, 4 and 5.

Proposition 3. *The following asymptotic holds under the same context as Lemma 10.*

$$-N\sum_{j=m_N}^{N-1} V_{\tau(j)}(r_{\tau(j)}) = -N^2 I_Q[\mu_Q] + NU_{\mu_Q}(0) + \frac{1}{6}\ln(r_1) + Nm_N q(0) + \frac{3}{4}m_N^2 - \frac{1}{2}\left(m_N^2 - m_N + \frac{1}{6}\right)\ln\left(\frac{4m_N}{NV_0^{(2)}(0)}\right) - \frac{m_N}{2} + O\left(N^{-\frac{1}{2}(3-5\epsilon)}\right). \tag{6.207}$$

Here, $I_Q[\mu_Q]$ *is as in Theorem 4,* ϵ *is as in Proposition 2 and*

$$2U_{\mu_Q}(0) := q(r_1) - 2\ln(r_1) - q(0). \tag{6.208}$$

Proof. We proceed as before and use the Euler-Maclaurin formula to write

$$\sum_{j=m_N}^{N-1} V_{\tau(j)}(r_{\tau(j)}) = \int_{m_N}^{N} V_{\tau(t)}(r_{\tau(t)})\mathrm{d}t - \frac{1}{2}\left(V_{\tau(N)}(r_{\tau(N)}) - V_{\tau(m_N)}(r_{\tau(m_N)})\right) + \frac{1}{12}\left[\partial_t V_{\tau(t)}(r_{\tau(t)})|_{t=N} - \partial_t V_{\tau(t)}(r_{\tau(t)})|_{t=m_N}\right] + O(N^{-3}). \tag{6.209}$$

Then, we are able to rewrite, setting $s = r_{\tau(j)}$,

$$\int_{m_N}^{N} V_{\tau(t)}(r_{\tau(t)})\mathrm{d}t = \int_{m_N}^{N}(q(r_{\tau(t)}) - 2\tau(t)\ln(r_{\tau(t)}))\mathrm{d}t, \tag{6.210}$$

$$= \frac{1}{2}\int_{r_{\tau(m_N)}}^{r_1}(q(s) - sq'(s)\ln(s))sV^{(2)}(s)N\mathrm{d}s, \tag{6.211}$$

$$= N\left[\frac{1}{2}\int_{r_{\tau(m_N)}}^{r_1} q(s)sV^{(2)}(s)\mathrm{d}s - \frac{1}{2}\int_{r_{\tau(m_N)}}^{r_1} s^2q'(s)\ln(s)V^{(2)}(s)\mathrm{d}s\right]. \tag{6.212}$$

Using that $r_1q'(r_1) = 2$, apply integration by parts on the second term and get

$$\frac{1}{2}\int_{r_{\tau(m_N)}}^{r_1} s^2q'(s)\ln(s)V^{(2)}(s)\mathrm{d}s \tag{6.213}$$

$$= \frac{1}{2}\int_{r_{\tau(m_N)}}^{r_1} s^2q'(s)\ln(s)\frac{1}{s}(sq'(s))'\mathrm{d}s, \tag{6.214}$$

$$= \frac{1}{4}\int_{r_{\tau(m_N)}}^{r_1}((sq'(s))^2)'\ln(s)\mathrm{d}s, \tag{6.215}$$

$$= \frac{1}{4}(sq'(s))^2\ln(s)|_{r_{\tau(m_N)}}^{r_1} - \frac{1}{4}\int_{r_{\tau(m_N)}}^{r_1}(sq'(s))^2\frac{1}{s}\mathrm{d}s, \tag{6.216}$$

$$= \ln(r_1) - \frac{1}{4}\int_{r_{\tau(m_N)}}^{r_1} sq'(s)^2\mathrm{d}s - \frac{m_N^2}{N^2}\ln(r_{\tau(m_N)}), \tag{6.217}$$

$$= \ln(r_1) - \frac{1}{4}(sq'(s)q(s))|_{r_{\tau(m_N)}}^{r_1} + \frac{1}{4}\int_{r_{\tau(m_N)}}^{r_1} q(s)(sq'(s))'\mathrm{d}s - \frac{m_N^2}{N^2}\ln(r_{\tau(m_N)}), \tag{6.218}$$

$$= \ln(r_1) - \frac{1}{2}q(r_1) + \frac{1}{4}\int_{r_{\tau(m_N)}}^{r_1} q(s)(sq'(s))'\mathrm{d}s - \frac{m_N^2}{N^2}\ln(r_{\tau(m_N)}) + \frac{m_N}{2N}q(r_{\tau(m_N)}). \tag{6.219}$$

Leaving us with

$$N\left[\frac{1}{4}\int_{r_{\tau(m_N)}}^{r_1} q(s)sV^{(2)}(s)\mathrm{d}s - \ln(r_1) + \frac{1}{2}q(r_1) - \frac{m_N^2}{N^2}\ln(r_{\tau(m_N)}) + \frac{m_N}{2N}q(r_{\tau(m_N)})\right] \tag{6.220}$$

$$:= NI_Q[\mu_Q] - \frac{1}{4}\int_{r_0}^{r_{\tau(m_N)}} q(s)sV^{(2)}(s)\mathrm{d}s - \frac{m_N^2}{N}\ln(r_{\tau(m_N)}) + \frac{m_N}{2}q(r_{\tau(m_N)}). \tag{6.221}$$

where we can expand some terms using the following results given by Proposition 2

$$\ln(r_{\tau(m_N)}) = \frac{1}{2}\ln\left(\frac{4\tau(m_N)}{V_0^{(2)}(0)}\right) + O(\tau(m_N)^{\frac{1}{2}}) = \frac{1}{2}\ln\left(\frac{4\tau(m_N)}{V_0^{(2)}(0)}\right) + O(N^{-\frac{1}{2}(1-\epsilon)}), \tag{6.222}$$

$$q(r_{\tau(m_N)}) = q(0) + \frac{1}{2}q''(0)(r_{\tau(m_N)})^2 + O((\tau(m_N)^{\frac{3}{2}})) = q(0) + \tau(m_N) + O(N^{-\frac{3}{2}(1-\epsilon)}), \quad (6.223)$$

and, using integration by parts we can solve the integral

$$\frac{1}{4}\int_{r_0}^{r_{\tau(m_N)}} q(s)sV^{(2)}(s)\mathrm{d}s = \frac{1}{4}\int_{r_0}^{r_{\tau(m_N)}} q(s)(sq'(s))'\mathrm{d}s, \quad (6.224)$$

$$= \frac{1}{2}\tau(m_N)q(r_{\tau(m_N)}) - \frac{1}{4}\int_0^{r_{\tau(m_N)}} s(q'(s))^2\mathrm{d}s, \quad (6.225)$$

$$= \frac{1}{2}\tau(m_N)q(r_{\tau(m_N)}) - \frac{1}{4}(q''(0))^2\int_0^{r_{\tau(m_N)}} s^3\mathrm{d}s + O(r^5_{\tau(m_N)}), \quad (6.226)$$

$$= \frac{1}{2}\tau(m_N)q(r_{\tau(m_N)}) - \frac{1}{4}\tau(m_N)^2 + O(N^{-5/2(1-\epsilon)}). \quad (6.227)$$

Therefore, for this first term, gathering the above and expanding using Proposition 2 and Equation 6.223,

$$\begin{aligned}\int_{m_N}^{N} V_{\tau(t)}(r_{\tau(t)})\mathrm{d}t = NI_Q[\mu_Q] - \frac{1}{2}\tau(m_N)q(r_{\tau(m_N)}) - \frac{1}{4}\tau(m_N)^2 + O(N^{-5/2(1-\epsilon)})\\ -\frac{m_N^2}{2N}\ln\left(\frac{4\tau(m_N)}{V_0^{(2)}(0)}\right) + O(N^{-\frac{3}{2}(1-\epsilon)}) + \frac{m_N}{2}\left(q(0)+\tau(m_N)\right) + O(N^{-\frac{1}{2}(1-\epsilon)})\end{aligned}, \quad (6.228)$$

$$= NI_Q[\mu_Q] - m_Nq(0) - \frac{3}{4}\frac{m_N^2}{N} + \frac{m_N^2}{2N}\ln\left(\frac{4m_N}{NV_0^{(2)}(0)}\right) + O\left(N^{-\frac{1}{2}(3-5\epsilon)}\right). \quad (6.229)$$

For the other terms in the Euler-Maclaurin formula we have

$$V_{\tau(N)}(r_{\tau(N)}) - V_{\tau(m_N)}(r_{\tau(m_N)}) = q(r_1) - 2\ln(r_1) - q(r_{\tau(m_N)}) + 2\tau(m_N)\ln(r_{\tau(m_N)}), \quad (6.230)$$

$$= q(r_1) - 2\ln(r_1) - q(0) - \frac{m_N}{N} + \frac{m_N}{N}\ln\left(\frac{4m_N}{NV_0^{(2)}(0)}\right) + O(N^{-3/2(1-\epsilon)}), \quad (6.231)$$

$$:= 2U_{\mu_Q}(0) - \frac{m_N}{N} + \frac{m_N}{N}\ln\left(\frac{4m_N}{NV_0^{(2)}(0)}\right) + O(N^{-3/2(1-\epsilon)}), \quad (6.232)$$

and,

$$\partial_t(V_{\tau(t)}(r_{\tau(t)}))|_{t=N} - \partial_t(V_{\tau(t)}(r_{\tau(t)}))|_{t=m_N} = \frac{2}{N}(\ln(r_{\tau(m_N)}) - \ln(r_1)), \quad (6.233)$$

$$= \frac{1}{N}\ln\left(\frac{4m_N}{NV_0^{(2)}(0)}\right) - \frac{2}{N}\ln(r_1) + O(N^{-1/2(3-\epsilon)}). \quad (6.234)$$

Finally, gathering terms, we get the lemma. □

Proposition 4. *We have following asymptotic*

$$\sum_{j=m_N}^{N-1} \ln V^{(2)}_{\tau(j)}(r_{\tau(j)}) = NE_Q(\mu_Q) - \frac{1}{2}\ln\left(\frac{V^{(2)}_1(r_1)}{V^{(2)}_0(r_0)}\right) - m_N \ln\left(V^{(2)}_0(0)\right) + O(N^{-1/2(1-\epsilon)}), \tag{6.235}$$

where $E_Q[\mu_Q]$ is as in Theorem 4.

Proof. Again, we start with the Euler-Maclaurin formula

$$\sum_{j=m_N}^{N-1} \ln V^{(2)}_{\tau(j)}(r_{\tau(j)}) = \int_{m_N}^{N} \ln V^{(2)}_{\tau(t)}(r_{\tau(t)})\mathrm{d}t - \frac{1}{2}\ln\left(\frac{V^{(2)}_1(r_1)}{V^{(2)}_{\tau(m_N)}(r_{\tau(m_N)})}\right) + \frac{1}{12}\left[\partial_t \ln V^{(2)}_{\tau(t)}(r_{\tau(t)})|_{t=N} - \partial_t \ln V^{(2)}_{\tau(t)}(r_{\tau(t)})|_{t=m_N}\right] + O(N^{-3}). \tag{6.236}$$

We first tackle the integral on the right hand side. Changing variables $s = r_{\tau(t)}$ we have

$$\int_{m_N}^{N} \ln V^{(2)}_{\tau(t)}(r_{\tau(t)})\mathrm{d}t = \frac{N}{2}\int_{r_{\tau(m_N)}}^{r_1} \ln(V^{(2)}(s))(sq'(s))'\mathrm{d}s, \tag{6.237}$$

$$= \frac{N}{2}\int_{r_{\tau(m_N)}}^{r_1} \ln(V^{(2)}(s))sV^{(2)}(s)\mathrm{d}s, \tag{6.238}$$

$$= \frac{N}{2}\left[\int_{r_0}^{r_1} \ln(V^{(2)}(s))sV^{(2)}(s)\mathrm{d}s - \int_{r_0}^{r_{\tau(m_N)}} \ln(V^{(2)}(s))sV^{(2)}(s)\mathrm{d}s\right], \tag{6.239}$$

$$= \frac{N}{2}\left[2E_Q[\mu_Q] - \int_{r_0}^{r_{\tau(m_N)}} \ln(V^{(2)}(s))sV^{(2)}(s)\mathrm{d}s\right], \tag{6.240}$$

where one can also compute

$$\int_{r_0}^{r_{\tau(m_N)}} \ln(V^{(2)}(s))sV^{(2)}(s)\mathrm{d}s \tag{6.241}$$

$$= \int_{r_0}^{r_{\tau(m_N)}} \ln\left(V^{(2)}_0(0) + V^{(2)}_0(0)s + O(s^2)\right) s\left(V^{(2)}_0(0) + V^{(2)}_0(0)s + O(s^2)\right)\mathrm{d}s, \tag{6.242}$$

$$= \int_{r_0}^{r_{\tau(m_N)}} \ln\left(V^{(2)}_0(0)\right) sV^{(2)}_0(0)\mathrm{d}s + O(r^3_{\tau(m_N)}), \tag{6.243}$$

$$= \ln\left(V^{(2)}_0(0)\right) r^2_{\tau(m_N)} V^{(2)}_0(0) + O(r^3_{\tau(m_N)}), \tag{6.244}$$

$$= \frac{2m_N}{N}\ln\left(V^{(2)}_0(0)\right) + O(N^{-3(1-\epsilon)}) \tag{6.245}$$

noting the definition for V and Proposition 2. Furthermore, we know

$$\partial_t \ln V^{(2)}_{\tau(t)}(r_{\tau(t)}) = \partial_V\left(\ln V^{(2)}_{\tau(t)}(r_{\tau(t)})\right)\partial_r\left(V^{(2)}_{\tau(t)}(r_{\tau(t)})\right)\partial_\tau\left(r_{\tau(t)}\right), \tag{6.246}$$

$$= \frac{2\partial_r V^{(2)}(r_{\tau(t)})}{Nr_{\tau(t)}\left(V^{(2)}_{\tau(t)}(r_{\tau(t)})\right)^2}, \tag{6.247}$$

$$= O\left(N/m_N\right)^{-1/2}. \tag{6.248}$$

Therefore, for the third term on the right hand side

$$\partial_t \ln V^{(2)}_{\tau(t)}(r_{\tau(t)})|_{t=N} - \partial_t \ln V^{(2)}_{\tau(t)}(r_{\tau(t)})|_{t=m_N} = \frac{2\partial_r V^{(2)}(r_1)}{N r_1 \left(V^{(2)}_1(r_1)\right)^2} - O\,(N/m_N)^{-1/2}, \tag{6.249}$$

$$= O\,(N/m_N)^{-1/2} = O(N^{-1/2(1-\epsilon)}). \tag{6.250}$$

Then, simply expanding the series on $V^{(2)}_{\tau(m_N)}(r_{\tau(m_N)})$, we can write,

$$\sum_{j=0}^{N-1} \ln V^{(2)}_{\tau(j)}(r_{\tau(j)}) = N E_Q(\mu_Q) - \frac{1}{2}\ln\left(\frac{V^{(2)}_1(r_1)}{V^{(2)}_0(0)}\right) - m_N \ln\left(V^{(2)}_0(0)\right) + O(N^{-1/2(1-\epsilon)}). \tag{6.251}$$

□

Proposition 5. *The following asymptotic holds under the same context as Lemma 10.*

$$\sum_{j=m_N}^{N-1} \ln(r_{\tau(j)}) = N\left[-U_{\mu_Q}(0) + \frac{1}{2}\tau(m_N)\right] - \frac{1}{2}\ln(r_1) - \frac{2m_N-1}{4}\ln\left(\frac{4\tau(m_N)}{V^{(2)}_0(0)}\right) + O(N^{-\epsilon}) \tag{6.252}$$

where $U_{\mu_Q}(0)$ is as in Proposition 3.

Proof. As usual, we start with the Euler-Maclaurin formula.

$$\sum_{j=m_N}^{N-1} \ln(r_{\tau(m_N)}) = \int_{m_N}^{N} \ln(r_{\tau(t)})\mathrm{d}t - \frac{1}{2}\ln\left(\frac{r_1}{r_{\tau(m_N)}}\right) + \frac{1}{12}\left[\partial_t \ln r_{\tau(t)}|_{t=N} - \partial_t \ln r_{\tau(t)}|_{t=m_N}\right] \tag{6.253}$$

where we can directly note that

$$\partial_t \ln r_{\tau(t)}|_{t=m_N} = \partial_r \ln r_{\tau(t)} \partial_\tau r_{\tau(t)} \partial_t \tau(t)|_{t=m_N} = \left.\frac{2}{r^2_{\tau(t)} V^{(2)}_{\tau(t)}(r_{\tau(t)}) N}\right|_{t=m_N} = O(m_N^{-1}), \tag{6.254}$$

$$\partial_t \ln r_{\tau(t)}|_{t=N} = \left.\frac{2}{r^2_{\tau(t)} V^{(2)}_{\tau(t)}(r_{\tau(t)}) N}\right|_{t=N} = O(N^{-1}). \tag{6.255}$$

Now, for the other terms we start with the first term, the integral, to get using integration by parts

$$\int_{m_N}^{N} \ln(r_{\tau(t)})\mathrm{d}t = \frac{N}{2}\int_{r_{\tau(m_N)}}^{r_1} \ln(s)(s'q(s))'\mathrm{d}s, \tag{6.256}$$

$$= \frac{N}{2}\left[\ln(s)sq'(s)\Big|^{r_1}_{r_{\tau(m_N)}} - \int_{r_{\tau(m_N)}}^{r_1} q'(s)\mathrm{d}s\right], \tag{6.257}$$

$$= \frac{N}{2}\left[\ln(r_1)r_1q'(r_1) - \ln(r_{\tau(m_N)})r_{\tau(m_N)}q'(r_{\tau(m_N)}) - q(r_1) + q(r_{\tau(m_N)})\right], \tag{6.258}$$

$$= \frac{N}{2}\left[2\ln(r_1) - 2\tau(m_N)\ln(r_{\tau(m_N)}) - q(r_1) + q(r_{\tau(m_N)})\right]. \tag{6.259}$$

Ultimately, using Equation 6.222 and Equation 6.223,

$$\int_{m_N}^{N} \ln(r_{\tau(t)})\mathrm{d}t - \frac{1}{2}\ln\left(\frac{r_1}{r_{\tau(m_N)}}\right) + O(m_N^{-1}) = N\left[\ln(r_1) - \frac{q(r_1)-q(0)}{2} + \frac{1}{2}\tau(m_N)\right] - \frac{1}{2}\ln(r_1) - \frac{2m_N-1}{4}\ln\left(\frac{4\tau(m_N)}{V_0^{(2)}(0)}\right) + O(N^{-\epsilon}). \quad (6.260)$$

□

Finally, we turn our attention to gathering these propositions and proving the original lemma.

Proof. (Lemma 10) We start by stating from Equation 6.131, with the new limits,

$$\frac{1}{N}\sum_{j=m_N}^{N-1} \mathfrak{B}(r_{\tau(j)}) = -\frac{3}{32}\ln\left(\frac{V_1^{(2)}(r_1)}{V_{\tau(m_N)}^{(2)}(r_{\tau(m_N)})}\right) - \frac{1}{32}\left(\frac{r_1 V_1^{(3)}(r_1)}{V_1^{(2)}(r_1)} - \frac{r_{\tau(m_N)} V_{\tau(m_N)}^{(3)}(r_{\tau(m_N)})}{V_{\tau(m_N)}^{(2)}(r_{\tau(m_N)})}\right) - \frac{1}{12}\int_{r_{\tau(m_N)}}^{r_1} \frac{V^{(3)}(r)^2}{V^{(2)}(r)^2} r\mathrm{d}r + O(N^{-2}). \quad (6.261)$$

using Proposition 2 and Equation 6.222 we can rewrite

$$a \quad (6.262)$$

$$r_\tau = \left(\frac{2\tau}{q''(0)}\right)^{\frac{1}{2}} + O(\tau) = \left(\frac{4\tau}{V_0^{(2)}(0)}\right)^{\frac{1}{2}} + O(\tau) \quad (6.263)$$

$$\ln(r_{\tau(m_N)}) = \frac{1}{2}\ln\left(\frac{4\tau(m_N)}{V_0^{(2)}(0)}\right) + O(\tau(m_N)^{\frac{1}{2}}) = \frac{1}{2}\ln\left(\frac{4\tau(m_N)}{V_0^{(2)}(0)}\right) + O(N^{-\frac{1}{2}(1-\epsilon)}), \quad (6.264)$$

We should have by the asymptotic for h_j in Theorem 5 , referring to $j > m_N$

$$\sum_{j=m_N}^{N-1} \ln(h_j) = \sum_{j=m_N}^{N-1}\left(-NV_{\tau(j)}(r_{\tau(j)}) + \ln(r_{\tau(j)}) - \frac{1}{2}\ln V_{\tau(j)}^{(2)}(r_{\tau(j)}) + \frac{1}{N}\mathfrak{B}(r_{\tau(j)})\right) + \frac{(N-m_N)}{2}\ln\frac{8\pi}{N} + O(m_N^{-1/2}(\ln N)^\alpha) + O(N^{\frac{1}{2}(1-5\epsilon)}). \quad (6.265)$$

By the lemmas just proved, we can write as $N \to \infty$,

$$\sum_{j=m_N}^{N-1} \ln(h_j) = -N^2 I[\mu_Q] + \frac{(N-m_N)}{2}\ln\frac{8\pi}{N} + Nm_N q(0) + \frac{3}{4}m_N^2 - \frac{N}{2}E_Q[\mu_Q] - \frac{m_N}{2}(m_N+1)\ln\left(\frac{4}{V_0^{(2)}(0)}\right) - \frac{1}{2}\left(m_N^2 - \frac{1}{6}\right)\ln\left(\frac{m_N}{N}\right) + \frac{1}{6}\ln\left(\frac{4}{V_0^{(2)}(0)}\right) - \frac{1}{3}\ln(r_1) + \frac{5}{32}\ln\left(\frac{V_1^{(2)}(r_1)}{V_{\tau(m_N)}^{(2)}(r_{\tau(m_N)})}\right) - \frac{1}{32}\left(\frac{r_1 V_1^{(3)}(r_1)}{V_1^{(2)}(r_1)} - \frac{r_{\tau(m_N)} V_{\tau(m_N)}^{(3)}(r_{\tau(m_N)})}{V_{\tau(m_N)}^{(2)}(r_{\tau(m_N)})}\right) - \frac{1}{12}\int_{r_{\tau(m_N)}}^{r_1} \frac{V^{(3)}(r)^2}{V^{(2)}(r)^2} r\mathrm{d}r + O(N^{-12}(\ln N)^3). \quad (6.266)$$

We just rewrite

$$\sum_{j=m_N}^{N-1} \ln(h_j) = -N^2 I[\mu_Q] + \frac{(N-m_N)}{2} \ln\frac{8\pi}{N} + N m_N q(0) + \frac{3}{4} m_N^2 - \frac{N}{2} E_Q[\mu_Q]$$
$$- \frac{m_N}{2}(m_N+1)\ln\left(\frac{4}{V_0^{(2)}(0)}\right) - \frac{1}{2}\left(m_N^2 - \frac{1}{6}\right)\ln\left(\frac{m_N}{N}\right) + F_V[D_{r_1}] + O(N^{-12}(\ln N)^3). \quad (6.267)$$

where

$$F_V[D_{r_1}] := \frac{1}{6}\ln\left(\frac{4}{V_0^{(2)}(0)}\right) - \frac{1}{3}\ln(r_1) + \frac{5}{32}\ln\left(\frac{V_1^{(2)}(r_1)}{V_0^{(2)}(r_0)}\right) - \frac{1}{32}\frac{r_1 V_1^{(3)}(r_1)}{V_1^{(2)}(r_1)} - \frac{1}{12}\int_{r_0}^{r_1} \frac{V^{(3)}(r)^2}{V^{(2)}(r)^2} r \mathrm{d}r \quad (6.268)$$

□

And here finally we return to the full summation

Proof. (Theorem 7) Gathering Lemma 9 and Lemma 10, we get

$$\sum_{j=0}^{N-1} \ln(h_j) = -N^2 I[\mu_Q] - \frac{1}{2} N \ln(N) - \frac{N}{2}\left(E_Q[\mu_Q] - \ln(8\pi)\right) - \frac{\ln(N)}{12} + \zeta'(-1) + F_V[D_{r_1}] + O(N^{-12}(\ln N)^3). \quad (6.269)$$

All the terms depending on m_N cancels out. □

6.2.2.3 Free Energy Disk

We can finally prove Theorem 4.

Proof. (Theorem 4) From

$$\sum_{j=0}^{N} \ln(h_j) = -N^2 I_Q[\mu_Q] - \frac{1}{2} N \ln N - \frac{N}{2}\left(E_Q[\mu_Q] - \ln(8\pi)\right)$$
$$- \frac{\ln N}{12} + \zeta'(-1) + F_V[D_{r_1}] + O(N^{-1/12}(\ln N)^3) \quad (6.270)$$

We can get the asymptotic expansion for the Free energy noting that all termns involving m_N vanish and using that

$$\ln(N!) = N\ln(N) - N + \frac{1}{2}\ln(N) + \frac{1}{2}\ln(2\pi) + O(N^{-1}) \quad (6.271)$$

that is,

$$\ln \mathcal{Z}_N = \ln N! - \sum_{j=0}^{N-1} \ln h_j \tag{6.272}$$

$$= \begin{array}{l} -N^2 I_Q[\mu_Q] - \frac{1}{2} N \ln N + N\left(\frac{\ln(4\pi)}{2} - 1 - \frac{1}{2} E_Q[\mu_Q]\right) - \frac{5}{12}\ln N \\ + \frac{\ln(8\pi)}{2} + \frac{1}{2}\zeta'(-1) + \frac{1}{2} F_V[D_{r_1}] + \frac{5}{4}\ln(2) + \frac{1}{8}\ln\left(\frac{V_0^{(2)}(0)}{V_1^{(2)}(r_1)}\right) + O(N^{-1/12}(\ln N)^3) \end{array} \tag{6.273}$$

□

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